

Chapter III Research Design and Methodology

A. Hypothesis

The study trains and compresses the student model via three compression mechanisms: Logit Knowledge Distillation (introduced in E2), Channel-Wise Knowledge Distillation (added in E3), and INT8 Quantization-Aware Training (applied in E5 and E6), intending to provide a more favorable accuracy–efficiency–robustness trade-off than the lightweight student baseline evaluated through segmentation accuracy on the clean test set, CPU proxy efficiency benchmarking, and robustness on selected corruption settings. As contrastive ablation arms, the study includes E4 (PTQ from E1) and E7 (PTQ from E3), which enables a 2×2 ablation of PTQ versus QAT, with and without prior distillation (E4 vs E5, E7 vs E6). Using SegNeXt-B with the MSCAN-B backbone (Guo et al., 2022) as the teacher model, knowledge is transferred from the teacher to a MobileNetV3-Large with task-adapted LR-ASPP head (Howard et al., 2019) student, leveraging response-level logit guidance and channel-wise feature distribution alignment from the teacher (Hinton et al., 2015; Shu et al., 2021). Subsequently, INT8 QAT is applied to compress the distilled student for deployability: reduced model size and memory footprint are guaranteed by the format change, while runtime improvements depend on backend and hardware support and are reported through CPU proxy efficiency benchmarking only (Jacob et al., 2018; Krishnamoorthi, 2018). Overall, the aim is for the resulting model to show acceptable segmentation performance, reduced model size, potentially improved inference latency where supported by the selected backend, decreased memory footprint, and better robustness on the corrupted image test set, evaluated by mIoU-C and RPD.

In this proposed study, improvement is judged by a favorable trade-off across all three dimensions: segmentation accuracy, efficiency, and robustness, rather than peak performance on any single dimension. Each dimension is not expected to reach the peak in isolation; rather, it is considered improved when it offers an acceptable compromise in terms of all three dimensions. Segmentation accuracy will be evaluated based on mean IoU (mIoU) and Dice Coefficient (Müller et al., 2022; Ren et al., 2025), efficiency based on model size, FLOPs, latency, and peak memory usage, and robustness through corrupted mIoU (mIoU-C) and Relative Performance Drop (RPD) on the corrupted image test set (Hendrycks & Dietterich, 2019; Kamann & Rother, 2020). These criteria are compared against the appropriate baseline or aligned comparison stage as defined in Table 3.5. Where suitable, pairwise inferences for per-image accuracy and robustness metrics will be applied as detailed in Section F. Efficiency metrics, particularly deterministic measures such as model size, parameter count, and FLOPs, will be evaluated based on descriptive comparison.

The general null hypothesis of the study is as follows:

H₀: The proposed pipeline, comprised of Logit Knowledge Distillation, Channel-Wise Knowledge Distillation, and INT8 Quantization-Aware Training, fails to outperform the baseline FP32 student (E1) and the alternative ablation configurations (E2–E5, E7) on the planned pairwise comparisons in terms of the accuracy and robustness of the MobileNetV3-based student on plant lesion segmentation, nor does it provide a more favorable accuracy–efficiency–robustness trade-off under resource-constrained conditions.

Equivalently, **H₀** decomposes into a family of pairwise null hypotheses $H_{0,k}$ ($k = 1 \dots 8$) corresponding to the eight planned tests in Table 3.6; family-wise Type I error is controlled at $\alpha = 0.05$ via Holm-Bonferroni correction (Section F.1.d).

H₁: The proposed pipeline utilizing Logit Knowledge Distillation, Channel-Wise Knowledge Distillation, and INT8 Quantization-Aware Training improves segmentation accuracy and robustness of the MobileNetV3-based student on plant lesion segmentation relative to the baseline FP32 student (E1) and the alternative ablation configurations (E2–E5, E7) and yields the most favorable accuracy–efficiency–robustness trade-off among the configurations evaluated in this study (E1–E7), with the SegNeXt-B teacher as a descriptive upper-bound reference only.

Published results on the PlantSeg benchmark dataset by Wei et al. (2026) will only be considered as reference material for contextual comparisons and are not treated as inferential comparators because Wei et al. (2026) per-image scores are not available for paired testing, and unmatched preprocessing, training procedure, and evaluation conditions would in any case invalidate any statistical comparison.

In more detail, the research questions and associated sub-hypotheses for experimental conditions E1 through E7 are as follows:

H_{1a}: Student trained with Logit Knowledge Distillation (E2) provides higher per-image mIoU than the FP32 student baseline (E1), measured by per-image mIoU (the unit of paired inferential testing, Section F.1.b). Dice Coefficient is reported descriptively at the dataset level only and is not separately tested. This hypothesis will be tested using a one-tailed Wilcoxon signed-rank test on per-image $\Delta\text{mIoU} = \text{mIoU}(\text{E2}) - \text{mIoU}(\text{E1})$, where $\Delta\text{mIoU} > 0$ indicates improvement.

H_{1b}: Student trained with Logit Knowledge Distillation and Channel-Wise Knowledge Distillation (added in E3) yields higher per-image mIoU than the Logit KD-only student (E2).

This is expected because Channel-Wise Knowledge Distillation provides dense, feature-level channel guidance that supplements the response-level probability guidance of Logit KD alone (Hinton et al., 2015; Shu et al., 2021). This hypothesis will be tested using a one-tailed Wilcoxon signed-rank test on per-image $\Delta\text{mIoU} = \text{mIoU}(\text{E3}) - \text{mIoU}(\text{E2})$, where $\Delta\text{mIoU} > 0$ indicates improvement.

H_{1c}: INT8 QAT yields higher per-image mIoU than INT8 Post-Training Quantization (PTQ) when each quantization technique originates from the same source checkpoint, tested by two paired comparisons: E4 vs E5 (undistilled) and E7 vs E6 (distilled), both contributing to the family-wise Holm-Bonferroni correction over all 8 inferential tests. Descriptive efficiency differences across all four quantization configurations (E4, E5, E6, E7), including model size, FLOPs/MACs, CPU proxy latency, and peak memory, are reported as supplementary observations and are not subject to inferential testing (Jacob et al., 2018; Krishnamoorthi, 2018). This hypothesis will be tested using two separate one-tailed Wilcoxon signed-rank tests on per-image ΔmIoU . Improvement is indicated when $\Delta\text{mIoU} > 0$ for the following comparisons: (i) $\Delta\text{mIoU} = \text{mIoU}(\text{E5}) - \text{mIoU}(\text{E4})$; and (ii) $\Delta\text{mIoU} = \text{mIoU}(\text{E6}) - \text{mIoU}(\text{E7})$.

The most favorable configuration for E6 is defined by retaining good results across segmentation accuracy, resource-constrained efficiency, and robustness against selected corruptions, with no single dimension required to reach its peak. For E6, the clean-test mIoU retained relative to the distilled FP32 student (E3) is evaluated against a 2.0 pp INT8 retention reference established in earlier works on classification models, complemented by the sub-one-percentage-point mIoU retention reported for per-channel INT8 segmentation models (Jacob et al., 2018; Krishnamoorthi, 2018; Nagel et al., 2019). In this study, which includes 115 disease classes with severe imbalance between classes and small lesion regions, the task is much harder than those classification benchmarks, so rather than applying a hard

pass/fail on the E3 to E6 drop, we interpret it as a factor to weigh relative to E3's actual accuracy level. This retention criterion is formally assessed as a one-sided non-inferiority check on dataset-level mIoU: E6 is considered non-inferior to E3 if the lower bound of the paired BCa 95% confidence interval ($B = 10,000$) for $\text{mIoU}(\text{E6}) - \text{mIoU}(\text{E3})$ exceeds -2.0 percentage points. This check is reported separately from the eight-test Holm-Bonferroni superiority family.

For E6, retention is not the only concern. Model size and CPU proxy latency are also checked against all FP32 student variants (E1, E2, and E3), and robustness against the FP32 baseline (E1) on corrupted images is assessed inferentially through the E1 vs. E6 mIoU-C comparison (Table 3.6), while efficiency metrics are reported descriptively. There are no absolute thresholds; E6 is considered most favorable when it achieves the best overall balance of accuracy retention, a smaller and faster model, and maintained quality on distorted images, among E1, E2, E3, E4, E5, and E7.

Hold: The full proposed pipeline, $\text{E6} = \text{Logit KD} + \text{CWD} + \text{INT8 QAT}$, yields the most favorable accuracy, efficiency, and robustness trade-off among all student configurations, E1 to E5 and E7. Operationally, this is supported when all four of the following hold: (i) accuracy retention, where the $\text{E3} \rightarrow \text{E6}$ drop in clean-test mIoU is small in absolute terms, interpreted against the 2.0 pp INT8 reference from classification benchmarks and the sub-one-percentage-point mIoU retention reported for per-channel INT8 segmentation models (Jacob et al., 2018; Krishnamoorthi, 2018; Nagel et al., 2019) with proportional allowance for PlantSeg's greater task difficulty, and assessed formally through the E3 vs E6 dataset-level mIoU non-inferiority check; (ii) accuracy improvement over baseline, where per-image $\text{mIoU}(\text{E6})$ is higher than per-image $\text{mIoU}(\text{E1})$, tested through the E1 vs. E6 comparison; (iii) robustness improvement over baseline, where per-image $\text{mIoU-C}(\text{E6})$ is

higher than per-image mIoU-C(E1), tested through the E1 vs. E6 comparison on mIoU-C; and (iv) efficiency, where the model size and CPU proxy latency of E6 are descriptively lower than all FP32 student variants, E1, E2, and E3. The SegNeXt-B teacher serves only as a descriptive upper-bound reference and is not subject to any inferential comparison.

B. Research Methods

1. Experimental and scientific method

The primary research methodology applied is the experimental and scientific method. This approach is most appropriate for the thesis as it involves constructing and validating a controlled experiment. This experiment is designed to address the research hypotheses H_{1a} - H_{1d} : whether Logit Knowledge Distillation, Channel-Wise Knowledge Distillation, and INT8 Quantization-Aware Training, applied jointly or in isolation, improve a lightweight semantic segmentation model on the accuracy–efficiency–robustness trade-off. Instead of simply describing how such a model functions or how knowledge distillation or INT8 quantization works in isolation, this thesis aims to rigorously assess the impact of combining these techniques on a practical problem (Pineau et al., 2021). The research involves designing experiments to evaluate architectures, prepare and document a dataset, systematically train a variety of model variants (representing distinct ablation stages), compress them, and compare their effectiveness based on a set of key metrics (Kerssies et al., 2024). The corrupted-dataset evaluation follows robustness benchmarking for common image corruptions and semantic segmentation (Hendrycks & Dietterich, 2019; Kamann & Rother, 2020). Instead of conducting separate experiments for each specific training stage, the same pipeline is run with varying subsets of components enabled or disabled. For example, E2 enables Logit KD only; E3 enables Logit KD and Channel-Wise KD. These configurations permit a clear

comparison against a baseline in each stage. By comparing each experimental stage against a meaningful baseline or aligned comparison stage, this approach aims to isolate the specific contributions of each component, thereby evaluating sub-hypotheses H_{1a} – H_{1d} .

2. Descriptive and statistical analysis method

Descriptive statistics summarize per-image and dataset-level metrics for all seven student stages (E1–E7) and the SegNeXt-B teacher reference across accuracy (mIoU, Dice, mAcc), robustness (mIoU-C, RPD, and rCD; evaluated for E1 through E7, with the SegNeXt-B teacher reported for mIoU-C and RPD only), and efficiency (model size, parameter count, FLOPs, CPU proxy latency, peak memory). Inferential pairwise testing on per-image mIoU and per-image mIoU-C is detailed in Section F (Rainio et al., 2024; Nariya et al., 2023).

This experimental structure naturally lends itself to comparative ablation analysis. Here, each of the seven configurations will be examined in relation to the other relevant configurations rather than only as an isolated sequential stage. More specifically, E2 versus E3 comparison highlights the effect of added Channel-Wise KD on the segmentation outcome over just Logit KD, supporting H_{1b} . Because both E2 and E3 are trained with the frozen SegNeXt-B teacher in the loop, the dominant per-iteration cost added relative to the undistilled E1 baseline — the teacher forward pass — is incurred identically by both stages. The only computational difference between E2 and E3 is the channel-wise distillation loss together with a training-only 1×1 projection head at the student's C5 stage, both of which are removed before evaluation. The E2-versus-E3 comparison is therefore approximately compute-matched by construction at the shared 80,000-iteration budget, unlike the E1-versus-E2 and E1-versus-E3 comparisons, in which the teacher forward pass makes the distilled stages more compute-intensive per iteration than the baseline. This distinction is

stated so that an improvement attributed to Channel-Wise KD in the E2-versus-E3 comparison reflects the added distillation signal rather than a difference in training compute, subject to the per-stage compute logs reported in Chapter 4. E4 versus E5 compares PTQ with QAT before distillation on an undistilled student model (E1), while E7 versus E6 performs the same comparison on a distilled student model (E3). E4 versus E7 determines whether distillation improves the performance of PTQ, and E5 versus E6 checks whether distillation improves the performance of QAT. The study uses a comparative ablation analysis for clear attribution of observed effects to Logit KD alone versus Logit KD plus Channel-Wise KD, as well as to INT8 compression with PTQ and QAT before and after distillation, considering their accuracy retention, robustness, and descriptive efficiency impacts. Additional comparisons, including E1 vs. E2 (H_{1a}), E1 vs. E3 (supporting H_{1a} and H_{1b}), E3 vs. E6 (accuracy retention, H_{1d}), and E1 vs. E6 (main pipeline evaluation on both clean accuracy and corrupted mIoU-C, H_{1d}), complete the inferential set of eight tests under Holm-Bonferroni correction.

C. Research Design

The study employs a research design classified as experimental, comparative, and model-development oriented. This research design was adopted as the goal of the thesis is to design and evaluate a lightweight yet robust semantic segmentation pipeline instead of simply describing the existing ones. It is experimental in the sense that factors are altered under controlled conditions in model training and compression schemes. It is model-development oriented, as the proposed final model will be developed by means of implementation, optimization, compression, as well as a stage for verification. Nevertheless, this should not be mistaken as a genuine developmental research method in terms of human respondents, observation over a period of time, and similar conditions. This type of research design

matches that of a computational benchmarking setup where machine learning experiments should be executed in identical datasets, metrics, as well as reporting format (Weber et al., 2019; Pineau et al., 2021).

The design comprises two distinct phases. The preliminary phase consists of essential setup procedures before starting the actual seven-stage experimentation cycle, which is denoted by E1 through E7. The experimental cycle focuses on developing, training, and validating the proposed model across several compression techniques and multiple stages of evaluation.

This study will address four core research objectives. Objective 1 will be addressed through Experiment E1. In E1, the standard FP32 MobileNetV3-based student, retaining native Hard-Swish/ReLU activations through the quantizable TorchVision implementation, with a task-adapted Lite-ASPP/LR-ASPP-style head is trained directly on the PlantSeg dataset. This forms the initial baseline to gauge the model's performance without any distillation or compression techniques. Objective 2 investigates the effect of teacher guidance on the lightweight student model's performance on clean test images. Experiment E2 involves training the student model using Logit Knowledge Distillation. Experiment E3 combines Logit Knowledge Distillation with Channel-Wise Knowledge Distillation (CWD) to see if feature-level guidance further improves segmentation results compared to E2. Objective 3 is concerned with the efficiency of resource-constrained models by evaluating model size, Floating Point Operations (FLOPs), CPU proxy latency, and peak memory usage. INT8 compression is achieved through Post-Training Quantization on E1, yielding E4. INT8 compression via Quantization-Aware Training on E1 results in E5. Post-Training Quantization on E3 leads to E7, while Quantization-Aware Training on E3 results in E6. FLOPs are a measure of computation, and QAT is not assumed to reduce theoretical FLOPs

unless profiler measurements indicate architecture-level changes or backend-specific behavior. Objective 4 is addressed in the clean-test and corrupted-test evaluation. Clean-test performance for all experimental stages (E1–E7) and the SegNeXt-B teacher reference is evaluated using the mean Intersection over Union (mIoU) and Dice Coefficient (Müller et al., 2022; Ren et al., 2025). Model robustness is tested against corrupted versions of the test images using corrupted mean IoU (mIoU-C) and the Relative Performance Drop (RPD) (Hendrycks & Dietterich, 2019; Kamann & Rother, 2020). E2 is additionally evaluated on corrupted data as a descriptive stage to decompose the corruption behavior of distillation into its Logit-KD (E1→E2) and CWD (E2→E3) contributions, and the SegNeXt-B teacher is evaluated on corrupted data as a descriptive reference providing an upper bound on absolute corrupted performance and a retention reference; the teacher is reported for mIoU-C and RPD only and is excluded from rCD and from all inferential testing.

All seven experimental stages (E1 through E7) and the SegNeXt-B teacher reference will be evaluated on the clean PlantSeg test set using mIoU and Dice Coefficient, serving primarily as checks that unacceptable drops in segmentation quality are not introduced during the compression stages (E4–E7) and robustness evaluation, using the 2.0 pp pre-registered tolerance.

1. Preliminary activities

Dataset acquisition and preparation: Images and lesion masks used throughout this work are taken from the official PlantSeg repository and subjected to a total of three types of image preparation processes. These preparation processes include: preprocessing, applied to the official training, validation and clean test sets, training augmentation, used exclusively with the training set, and corruption generation, applied only to clean test images for robustness testing purposes (Wei et al., 2026; Kerssies et al., 2024). During preprocessing, we verify

each image against its mask by confirming that they share the same file name or identifier, both are readable, the image dimensions are identical to the mask dimensions, and the mask pixels are within the permitted set of class indices. Unmatched, unreadable, dimension-mismatched or value-out-of-range images are reported and omitted. The study verifies that no image identifiers appear in more than one partition, confirming the integrity of data splits. These preliminary verification steps are mandatory and must be completed and documented before starting the training of E1. Verification logs are stored as JSON artifacts alongside each training run for reproducibility. Table 3.1 lists the entire checklist applied to the PlantSeg dataset.

Table 3.1 Dataset Preparation and Verification Checklist

Step	Procedure	Pass Criterion
Mask class count	Run <code>np.unique()</code> on all annotation PNGs from the downloaded dataset	Confirmed unique class count; model output layer defined accordingly
Image-mask pairing	Verify shared filename/ID, readability, and dimension match for every pair	Zero unmatched, unreadable, or dimension-mismatched pairs
Split integrity	Assert zero identifier overlap between train, val, and test partition ID sets	Overlap count = 0
Mask value range	Confirm all mask pixel values fall within the verified class set or equal 255	No out-of-range values present
Ignore-label	Confirm that value 255 is excluded from loss and metric computation	Per-term unit tests pass on a synthetic padded sample before training: cross-entropy, Dice, Logit KD, and CWD losses are unchanged when padded-pixel values are

		altered, and the CWD channel-wise spatial softmax sums to one over valid feature locations only. A teacher forward hook is verified to return the stride-16 MSCAN-B Stage-3 feature (320 channels) for a synthetic input before training, confirming the CWD alignment target.
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All data verification steps must be passed before initiating any model training or evaluation processes. Failure to pass any verification check is considered an error in data preparation and requires correction before proceeding. The number of output classes, identified in Step 1, dictates the number of output nodes in the student model's final classifier layer. The verified images and masks are then resized to a resolution of 512×512 . We apply bilinear interpolation to the images to preserve visual continuity, while masks are handled using nearest-neighbor interpolation to preserve integer class labels (Kerssies et al., 2024). Aspect-ratio-preserving padding introduces extra padding regions to image frames, which are assigned the integer value 255 in the corresponding segmentation mask. This value acts as an “ignore” label. These padding pixels are systematically excluded from both loss calculations during training and all metric calculations at evaluation time by setting `ignore_index = 255`, ensuring that they have no impact on the results. For cross-entropy, exclusion is handled natively through `ignore_index = 255`. For the Dice term, the validity mask is applied to both the predicted probabilities and the one-hot labels before the intersection and union are computed. For the Logit KD term, the per-pixel KL divergence is averaged over valid pixels only. We ensure the images are in RGB format, scale pixel values to the $[0, 1]$ range and then normalize the pixels per channel based on the ImageNet mean $[0.485, 0.456, 0.406]$ and

standard deviation [0.229, 0.224, 0.225] to match the pretrained initialization of MobileNetV3 and SegNeXt and finally cast the pixels to float32 in CHW format (Howard et al., 2019; Guo et al., 2022). The masks are cast to int64 in [H, W] format, storing integer class indices derived from the grayscale PlantSeg segmentations, with the class indexing kept consistent with the official data (Wei et al., 2026). Under the official PlantSeg label space, the plant-disease combinations are stored at grayscale indices 0 through 114, with the value 255 reserved as the ignore label for padded regions (Wei et al., 2026); the exact set of valid class indices, including whether a distinct background class is present, is confirmed empirically through the `np.unique()` mask inspection specified in Table 3.1, and the student's output layer is sized to this verified count. This 116-class count is treated as provisional until confirmed by the mask-value verification in Table 3.1; should the released annotations encode the background within the 1–115 index range rather than as a separate index, the output layer is resized to the empirically verified count before E1 training begins.

Training augmentation is applied solely to the training dataset and includes three main components: (1) the same image and mask transformations are used to handle geometric modifications; (2) image-only transformations are employed to account for color variations, specifically hue and saturation; and (3) class-aware inverse-frequency weighting is utilized to address imbalances in the frequencies of different disease categories (Yang et al., 2024; Keressies et al., 2024). For mask resizing during the crop operation, nearest-neighbor interpolation is used to preserve the integrity of the integer class labels. To compute class weights, the study calculates the square root of the ratio of total training pixels to the product of the number of classes and the number of pixels belonging to each class. This square-root inverse-frequency weighting compresses the weight range across the imbalanced classes to limit gradient instability under small batch sizes; the weights are normalized by the median weight, and the realized minimum, median, and maximum weights are reported rather than

imposing a fixed cap. These weights are applied to the cross-entropy term only; the Dice term is left unweighted because its per-class macro averaging already provides class-level balance. The class-frequency counts are computed over valid (non-255) training pixels only, so the ignore label is excluded from the weight computation. All other datasets, namely validation, clean test, and corrupted test sets, remain unaltered. Corruption is handled as a distinct process separate from training augmentation. Corruptions are applied to the resized uint8 RGB test images before padding and before tensor conversion and normalization, adhering to the physically accurate acquisition-order pipeline (Hendrycks & Dietterich, 2019; Kamann & Rother, 2020). Corruption is never applied to segmentation masks. Ground-truth masks for corrupted test images remain the same as those of the corresponding clean test images (Hendrycks & Dietterich, 2019; Kamann & Rother, 2020).

2. Core experimental pipeline: Ablation stages E1 to E7

Each stage introduces or isolates one component to attribute observed changes in accuracy, efficiency, or robustness to that specific intervention. The upper-limit benchmark comparison will be made using the Teacher Model in every case. Table 3.2 summarizes the six stages below:

The seven stages (E1 through E7) of the experimental pipeline will be carried out under the same experimental conditions for all comparisons so as to be valid to compare. Specifically, the same official PlantSeg splits, preprocessing pipeline, input image resolution, normalization parameters, and mask formatting will be used for every experimental condition throughout all stages. The environment, the set of metrics used, and the types and levels of corruption will also remain the same. All training runs use random seed 42 by default; if multi-seed validation is conducted for any stage (treated as optional due to compute constraints), all seeds used are reported in the appendix. All stages share the same

80,000-iteration training budget; the comparisons are therefore iteration-matched rather than wall-clock- or FLOPs-matched. Full compute-matched re-runs of every stage are beyond the compute budget of this study, so compute comparability is documented instead through the per-stage training-cost and GPU-hour logs and, for the E2-versus-E3 contrast specifically, through the shared frozen-teacher forward pass that makes that comparison approximately iso-compute by design. Where one stage is run at a single seed (for example, E2) while another is repeated across seeds (E1 and E3), the corresponding paired comparison rests on the single shared seed and the multi-seed variance of the repeated stage is not matched; this is acknowledged as a limitation on the strength of that paired comparison. If compute permits, a single extended-schedule logit-KD run (E2 trained to approximately 160,000 iterations at one seed) may be reported as a sanity check: should the channel-wise-distilled E3 at 80,000 iterations still exceed this longer-trained logit-KD baseline, the observed Channel-Wise KD gain cannot be attributed to under-training of the logit-KD baseline alone. This run is treated as an optional control, not a required stage of the locked E1–E7 design; if not run, it is recorded as future work. As a further optional control, the channel-wise feature-map loss weight may be varied around its default of 50 (for example, across $\{25, 50, 100\}$) at one seed, to confirm that the E3 improvement is stable across a small neighborhood of this hyper-parameter rather than dependent on a single setting; if not run, the fixed weight follows Shu et al. (2021) as a design choice, with sensitivity testing deferred to future work. In so doing, differences in data handling methods, input pipeline steps, or evaluation procedures will not be mistaken for improvements caused by the model compression pipeline (Kerssies et al., 2024).

Moreover, the data preprocessing details employed in this paper are standardized across all experiments listed in the paper, including every stage. This includes the comprehensive preprocessing chain used on all the original PlantSeg datasets (i.e., train,

validation, clean test sets), a set of augmentations only applicable to the training set, and a specific set of tests to check robustness on the clean and corrupted test images only. These steps are standardized to confirm that any changes noticed from E1 through E7 are due to the experiments performed in the stages rather than differences in data preprocessing methods or evaluation configuration (Kerssies et al., 2024).

The design builds up configuration complexity through a sequence of stages where one stage builds upon the other and controls for one element of improvement or another: first, E1 represents the baseline of the MobileNetV3 + LR-ASPP that is trained on PlantSeg with FP32 weights. Next, Experiment E2 incorporates Logit Knowledge Distillation, aiming to match the probability distributions of the student's output with the softened logits generated by the teacher model. Following E2, Experiment E3 builds upon this by combining Logit Knowledge Distillation with Channel-Wise Knowledge Distillation (CWD). CWD focuses on aligning the channel-wise activation distributions between the teacher's and the student's encoder features to provide denser, feature-level guidance. On the compression side, E4 begins from the E1 FP32 checkpoint and applies Post-Training Quantization (PTQ). Conversely, E5 starts from the same E1 FP32 checkpoint but utilizes Quantization-Aware Training (QAT). For E7, PTQ is applied to the E3 distilled checkpoint. For E6, QAT is applied to the E3 distilled checkpoint, representing the full pipeline developed in this study. Finally, in E6, the MobileNetV3-based student model, trained using both Logit and Channel-Wise Knowledge Distillation (E3), is further fine-tuned under INT8 QAT after the CWD projection head has been removed (Hinton et al., 2015; Shu et al., 2021). This represents the comprehensive integration of teacher-student knowledge transfer with INT8 compression (Jacob et al., 2018; Krishnamoorthi, 2018).

Table 3.2 Summary of Ablation Stages and Their Experimental Purpose

Stage	Configuration	Starting Checkpoint	Purpose	Main Comparison
Teacher	SegNeXt-B / MSCAN-B fine-tuned on PlantSeg	ADE20K-pretrained SegNeXt-B / MSCAN-B (MMSegmentation model zoo) fine-tuned on PlantSeg	Upper-bound segmentation reference. Not deployed.	Teacher vs. E6 (descriptive only)
E1	Student FP32 baseline	ImageNet-pretrained quantizable MobileNetV3-Large	Undistilled, unquantized anchor baseline	E1 vs. E2; E1 vs. E3; E1 vs. E6
E2	Student FP32 + Logit KD	Same ImageNet-pretrained MobileNetV3-Large initialization as E1; trained independently with Logit KD loss ($T_{\text{Logit}}=4$, λ_{logit} selected via validation sweep, reported in Chapter 4)	Tests whether response-level Logit KD improves over E1	E1 vs. E2
E3	Student FP32 + Logit KD + CWD	Same ImageNet-pretrained quantizable MobileNetV3-Large initialization as E1; trained with Logit KD + CWD loss ($T_{\text{Logit}}=4$, $T_{\text{CWD}}=4$,	Tests whether CWD adds improvement beyond Logit KD alone	E2 vs. E3

		$\alpha_{\text{CWD}}=50,$ $\beta_{\text{CWD}}=3$)		
E4	Student INT8 PTQ	E1 FP32 checkpoint	PTQ baseline on undistilled student; ~128 calibration images, per-channel weight quantization on depthwise convolutions	E4 vs. E5; E4 vs. E7
E5	Student INT8 QAT	E1 FP32 checkpoint	Tests whether QAT preserves accuracy better than PTQ on the undistilled student; fine-tuned from the E1 FP32 checkpoint with SGD lr 3×10^{-4} (cosine decay) for ~15 epochs, per-channel symmetric weight quantization on all convolutional layers, BN statistics frozen after 65–70% of training	E4 vs. E5; E5 vs. E6
E6	Logit KD + CWD + INT8 QAT	E3 checkpoint (projection head removed)	Full proposed pipeline. Initialized from the E3 distilled FP32 checkpoint with the CWD projection head removed, then fine-tuned with INT8 QAT under the same	E3 vs. E6; E7 vs. E6; E1 vs. E6; Teacher vs. E6

			configuration as E5	
E7	Student INT8 PTQ	E3 checkpoint (projection head removed)	PTQ on the distilled student. Initialized from the E3 checkpoint with the CWD projection head removed; same ~128 calibration images and per-channel-depth wise weight quantization as E4. Completes the 2×2 quantization $\times$ distillation matrix	E7 vs. E6; E4 vs. E7

All stages use the same PlantSeg official 70/10/20 split, preprocessing pipeline, input resolution (512×512), normalization parameters, and evaluation metrics. Differences in results across stages are attributed to the specific compression component introduced in each stage. The SegNeXt-B/MSCAN-B teacher is fine-tuned using the Wei et al. (2026) PlantSeg repository configuration that produced the published 42.05% benchmark: AdamW optimizer, learning rate 6e-5, weight decay 0.01, betas = (0.9, 0.999), decode-head lr_mult = 10, poly learning-rate schedule, 40,000 iterations, batch size 16, 512×512 crops, and cross-entropy loss. Reproduction is operationalized as recovering the benchmark within approximately ±1.5–2.0 mIoU percentage points rather than the exact single-run value, since Wei et al. (2026) report no seed or variance. Protocol matching, rather than maximal teacher accuracy, is the objective, because the teacher serves only as an upper-bound reference and any deviation would break comparability with the published benchmark.

E1: Student FP32 baseline. The MobileNetV3 together with LR-ASPP (Howard et al., 2019) will be trained independently on the PlantSeg dataset, without utilizing any Knowledge Distillation technique. Output stride 16 is obtained by applying atrous (dilated) convolution with dilation rate 2 in the final backbone stage (the depthwise convolutions of the C4–C5 region), with that stage's stride set to 1; the head itself contains no parallel atrous pyramid. The LR-ASPP head fuses two feature levels additively at the class-logit level rather than by concatenation: the high-level C5 branch (output stride 16) is processed by two parallel sub-branches — a 1×1 Conv-BN-ReLU producing 256 channels, and a large-kernel average-pooling branch followed by a 1×1 convolution, sigmoid, and bilinear upsampling — whose outputs are multiplied element-wise and bilinearly upsampled to output stride 8; the low-level skip feature (output stride 8) is projected by a 1×1 convolution to the 115 output channels, the upsampled high-level branch is projected by a separate 1×1 convolution to 115 channels, and the two are summed and upsampled to full resolution. Training configuration: SGD with momentum 0.9, initial learning rate $1e-2$, polynomial decay (power 0.9), weight decay $1e-4$, batch size 16, 80,000 iterations at 512×512 input. Validation runs every 4,000 iterations; the best-mIoU checkpoint on the validation split is retained. CE + Dice combined loss with class-aware inverse-frequency weighting. To make an assessment of how good the segmentation performance of the designed student can possibly get, the current E1 experimental condition serves as the baseline FP32 student for its architecture, without any knowledge gain or any form of compression, to estimate how well it can perform. All student stages are trained with the same supervised objective, the equally weighted sum of class-weighted cross-entropy and Dice loss, $L_{\text{sup}} = L_{\text{CE}} + L_{\text{Dice}}$ (Isensee et al., 2021). The FP32 student stages E1, E2, and E3 further share an identical training recipe — SGD with momentum 0.9, initial learning rate $1e-2$ with polynomial decay (power 0.9), weight decay $1e-4$, batch size 16, 80,000 iterations at 512×512 , validation every 4,000 iterations, and

best-validation-mIoU checkpoint selection — and differ only in the distillation terms added on top of L_{sup} , so that every difference among E1, E2, and E3 is attributable to the distillation objective rather than to training budget, optimizer, or checkpoint-selection differences. The Dice term is a soft Dice computed on predicted probabilities, aggregated per class across the batch and averaged over the classes present in the batch, with absent classes excluded from the average and a smoothing constant of $1e-5$ added to its numerator and denominator. During the distillation stages the distillation weights are linearly ramped from zero to their target values over the first training epoch, and global-norm gradient clipping is applied throughout. In all distillation stages the teacher is run in evaluation mode online and consumes the identical augmented input tensor used by the student, so that the teacher logits and stride-16 features correspond exactly to the student's view of each training sample.

E2: Student FP32 with Logit KD. The student is trained with standard response-based Knowledge Distillation using output distributions provided by the SegNeXt-B / MSCAN-B teacher model, against ground truths. This experimental condition examines whether or not KD offers any significant gains over E1, with just its plain response characteristics alone. To combine both response characteristics, Cross-Entropy and KL Divergence will be used between the output probabilities supplied from the teacher and student models, scaled by a temperature T . The temperature T is set to 4, consistent with the high-temperature soft-target regime of Hinton et al. (2015). The constant weight scaling the distillation term relative to the cross-entropy term, λ_{logit} , is not fixed a priori but is selected on the validation partition through a small pre-registered grid sweep over $\lambda_{\text{logit}} \in \{0.25, 0.5, 1, 2, 4\}$, a geometric grid centered on the conventional unit weight and bounded above near the locked Channel-Wise logit-map weight $\beta_{\text{CWD}} = 3$; because the temperature factor T^2 is already absorbed into the Logit KD term, λ_{logit} acts as an order-unity multiplier on a gradient-matched loss, which fixes the grid at this scale. The candidate maximizing dataset-level validation mIoU at the

default seed (42) is selected; ties falling within the validation-set noise band are resolved toward the smaller weight, and if the selected value lies at a grid boundary it is reported as such rather than the grid being extended. The held-out test partition is not consulted during selection, mirroring the selection-on-validation, reporting-on-test protocol already applied to the post-training-quantization calibration configuration; the selected value, together with the validation mIoU of every candidate, is reported in Chapter 4. This selected value of λ_{logit} is then reused unchanged in E3 so that the E2-versus-E3 comparison isolates the added Channel-Wise Distillation terms and is not confounded by a change in the Logit KD weight.

E3: Student FP32 with CWD. The student model will be trained using a combination of Logit Knowledge Distillation and Channel-Wise Knowledge Distillation (Shu et al., 2021). Channel-Wise Knowledge Distillation aligns the channel-wise activation distributions of the teacher and student models at the encoder feature level. This process involves treating each feature channel as an individual probability distribution over space and minimizing the Kullback-Leibler (KL) divergence between the corresponding teacher and student distributions (Shu et al., 2021). Following Shu et al. (2021), the channel-wise distillation term applies the T^2/C normalization, with $T = 4$ and C the teacher channel count at the aligned stride-16 stage (320 for MSCAN-B), and is added to the E2 objective with a weight of 50, giving the E3 loss $L_{\text{E3}} = L_{\text{CE}} + L_{\text{Dice}} + \lambda_{\text{logit}} \cdot L_{\text{LogitKD}} + 50 \cdot L_{\text{CWD_feat}} + 3 \cdot L_{\text{CWD_logit}}$. The ignore label is respected at the feature level by downsampling the validity mask to the stride-16 resolution and restricting the channel-wise spatial softmax and the KL divergence to valid, non-padded feature locations for both teacher and student. To enable CWD, a temporary 1×1 convolutional projection head is applied to the student's C5 encoder output during training only. Following Shu et al. (2021), CWD uses temperature $T_{\text{CWD}} = 4$, feature-map loss weight $\alpha_{\text{CWD}} = 50$ at the stride-16 C5 map, and logit-map loss weight $\beta_{\text{CWD}} = 3$. The 1×1 projection head maps the student's 160-channel C5 to the teacher's

matching channel dimension and is removed from the student before E6 and E7 quantization. This head adjusts the channel dimensions of the student's features to match those of the teacher before the CWD calculation. This projection head is removed from the student model before conducting experiments E6 and E7, which are derived from the E3 checkpoint. Removal is performed via `state_dict` editing before quantization observer insertion. E4 and E5 are derived from E1 and therefore do not contain the CWD projection head. The performance gap between E2 and E3 will be examined to determine if the inclusion of channel-wise feature-level guidance yields benefits that surpass those achieved solely through response-level Logit KD.

E4: Student INT8 PTQ. Post-Training Quantization will be applied onto the E1 baseline FP32 model, in a setup whereby E5 uses the same starting baseline to evaluate the contribution of the post-hoc quantization technique, without introducing another new model that'll further be manipulated. A calibration subset of 128 images sampled from the training partition (no augmentation, same preprocessing as the clean test set) is used to compute per-channel weight quantization parameters and histogram-based activation statistics (Krishnamoorthi, 2018). A subset of this size is sufficient because calibration estimates activation ranges rather than updating weights; Krishnamoorthi (2018) notes that on the order of a hundred mini-batches typically suffices for these ranges to converge. The identifiers of these calibration images are fixed and persisted as an explicit list, and the identical calibration subset is used for both E4 and E7 so that PTQ comparisons are not confounded by different calibration samples. Per-channel symmetric quantization is applied to all convolutional weight layers (per-channel depthwise handling is essential for MobileNet-class models per Krishnamoorthi, 2018); per-tensor asymmetric quantization is applied to activations. This E4 configuration provides a reference point for evaluating whether post-hoc quantization alone can preserve acceptable performance relative to the FP32 baseline and the QAT-based

alternative in E5. The post-training quantization baseline uses standard static INT8 quantization; more advanced post-training methods such as adaptive rounding (Nagel et al., 2020) and block reconstruction (Li et al., 2021) deliver their largest gains at sub-8-bit precision and fall outside the eager-mode toolchain used here, and are therefore not adopted at INT8.

E5: Student INT8 QAT. Following principles of INT8-only inference and quantization-aware training, Quantization-Aware Training will be deployed on top of the E1 baseline using PyTorch's `torch.ao.quantization` module with the QNNPACK backend (Jacob et al., 2018; Krishnamoorthi, 2018). To improve training stability, this will follow training of its original FP32 version (fine-tuning from the FP32 checkpoint). The quantization-aware fine-tuning uses stochastic gradient descent with momentum 0.9 and a base learning rate of 3×10^{-4} with cosine decay, recalibrated from the 1×10^{-5} ImageNet-scale value of Krishnamoorthi (2018) to account for the substantially smaller PlantSeg training partition and the correspondingly reduced step count. Fine-tuning is run for approximately 15 epochs with early stopping on validation mIoU. Because fine-tuning begins from a converged floating-point checkpoint with well-behaved activation statistics, activation quantization is enabled from the first step using moving-average range observers rather than delayed. Batch-normalization statistics are switched from batch to frozen long-term estimates after roughly 65–70% of training, and the range observers are frozen shortly afterward so that the remaining steps adapt the weights to fixed quantization parameters. The activation range is tracked with a moving-average observer, but no exponential moving average of the model weights is used, as averaged weights quantize less favorably than instantaneous weights (Krishnamoorthi, 2018); the final model is selected by best validation mIoU. Gradient clipping by global norm is applied for stability, and no distillation loss is active during quantization-aware fine-tuning; the conditions under which distillation is reinstated during

quantization are pre-registered as the E6-KD contingency. Conv–BN–ReLU sequences are fused via `torch.ao.quantization.fuse_modules` before observer insertion, while Hard-Swish and Hardsigmoid activations are quantized as standalone operators. EMA weight averaging is not used. Per-channel symmetric INT8 weight quantization is applied to all convolutional weight layers; per-tensor asymmetric UINT8 quantization is applied to activations. By contrasting E4 with E5, it can be determined whether QAT preserves accuracy better than PTQ methods. The quantization-aware training uses standard fake-quantization; learned-quantization methods such as LSQ (Esser et al., 2020) and PACT (Choi et al., 2018) improve on standard QAT chiefly at sub-8-bit precision and lack a direct export path to the deployment backend, and are therefore not adopted at INT8.

E6: Logit KD + CWD followed by INT8 QAT (full proposed pipeline) E6 is the full proposed pipeline. The student model checkpoint produced by Experiment E3 serves as the initial weight configuration for Experiment E6. Prior to commencing the Quantization-Aware Training process in E6, the CWD projection head is removed from the E3 checkpoint. This ensures that the final student model evaluated after quantization contains no layers exclusively utilized for training. The resulting student model is subsequently fine-tuned using INT8 QAT, employing the identical quantization configuration used in Experiment E5. Distillation is intentionally confined to the FP32 stage (E3) and is not reapplied during the E6 quantization-aware training phase; the distilled representation is carried into E6 solely through the initial weight configuration. This isolates the effect of INT8 quantization on the already-distilled student, which is precisely what the comparison between E3 and E6, and between E5 and E6, is intended to measure. Reapplying response-level distillation during quantization-aware training was considered but not adopted, as it would conflate the effect of quantization with that of continued teacher guidance and would obscure the accuracy-retention interpretation of the comparison between E3 and E6.

As a pre-registered contingency, a distillation-assisted variant of the proposed pipeline, denoted E6-KD, is specified in advance and is reported only if the supervised-only E6 stage fails to retain segmentation accuracy relative to the distilled FP32 student (E3). In E6-KD, the frozen SegNeXt-B / MSCAN-B teacher used in E3 is reinstated during quantization-aware fine-tuning, and the Logit KD and Channel-Wise Knowledge Distillation losses are reapplied at a reduced weight relative to their E3 values, while the supervised objective of class-weighted cross-entropy plus Dice is retained; the reduced distillation weight is used so that the soft-target regularization does not override the quantized student's adaptation to the INT8 grid (Kim et al., 2019). This contingency is motivated by evidence that teacher guidance applied during low-precision training can recover quantization-induced accuracy loss, with the recovery being most pronounced at lower bit-widths and for compact depthwise-separable backbones (Mishra & Marr, 2018; Kim et al., 2019; Zhao & Zhao, 2024). E6-KD is reported as an additive arm alongside E6 rather than as a replacement, so that the E3-versus-E6 comparison continues to isolate the effect of INT8 quantization, while the E6-versus-E6-KD comparison isolates the incremental effect of distillation applied during quantization-aware training

These E6 experiments would provide insight into whether a desirable balance in accuracy and robustness can be achieved while still using an INT8 version of the distilled student. This will then be measured against E3 to track accuracy and robustness retention and against E1 to gauge the benefits derived from implementing the proposed comprehensive technique.

E7: Student INT8 PTQ on distilled checkpoint. Post-Training Quantization is applied to the E3 distilled FP32 checkpoint (with the CWD projection head removed). The same ~128 calibration images and per-channel-depthwise weight quantization configuration used in E4 are applied here. E7 enables two comparisons not available without it: PTQ vs QAT on the

distilled student (E7 vs E6, H_{1c} extended), and the effect of distillation on the PTQ path (E4 vs E7, supporting H_{1d}).

3. Justification over alternative approaches

Table 3.3 Comparison of Alternative Research Design Approaches and Justification for Rejection

Alternative Approach	Reason for Exclusion
Plain lightweight student as the final deliverable (no KD, no quantization)	Would not test whether teacher-guided knowledge transfer or INT8 compression can improve the student. Retained internally as the E1 baseline against which all other stages are compared, but rejected as the proposed final pipeline.
Efficiency-only design without robustness testing	May produce a fast model whose performance under degraded image conditions has not been evaluated
Clean-set accuracy reporting only	Does not evaluate model behavior under selected synthetic image corruption conditions
PTQ as the only quantization technique	Documented to cause catastrophic accuracy loss on MobileNet-class models (e.g., MobileNet-v1 per-layer PTQ: 0.001 vs 0.709 FP32 per Krishnamoorthi, 2018). Retained as E4 (PTQ from E1) and E7 (PTQ from E3) to empirically compare PTQ against QAT and to complete the 2×2 quantization × distillation matrix; rejected as the only quantization technique.
Logit KD only	Transfers response-level teacher guidance through softened output logit matching but does not include channel-wise feature-level distillation; retained as E2 to isolate the contribution of response-level teacher guidance before evaluating the additional benefit of Channel-Wise Knowledge Distillation in E3. Channel-wise feature distillation (CWD; Shu et al., 2021) is specifically designed for dense

	prediction tasks and aligns spatial channel distributions in a way logit-level KD cannot. Because CWD operates on intermediate features, the E3-versus-E2 contrast also functions as a check on feature-level negative transfer under the teacher–student architecture mismatch. Rejected as the only KD method; retained as E2 to isolate Logit KD's marginal contribution before adding CWD in E3.
PTQ applied only to the undistilled student (E4) without a corresponding PTQ stage from the distilled student	Without E7, the study cannot compare PTQ versus QAT after distillation (E7 vs E6), or determine whether distillation improves the PTQ path (E4 vs E7). E7 is included to complete the 2×2 quantization comparison matrix and is computationally trivial as it requires only a calibration pass.
PlantVillage / Plant Pathology 2020 / FieldPlant as the benchmark dataset	These provide image-level disease labels or limited pixel-level annotation. PlantSeg (Wei et al., 2026) is the only public benchmark offering 115-class pixel-level lesion segmentation across 34 plant species. Rejected; PlantSeg adopted.
SegNeXt-L / MSCAN-L or ConvNeXt-L as the teacher	Both outperform SegNeXt-B on PlantSeg (44.52% and 46.24% mIoU respectively vs 42.05% for SegNeXt-B per Wei et al., 2026). Rejected for capacity/distillation-compatibility reasons: SegNeXt-L (49M params) and ConvNeXt-L (121M) are heavier teachers whose feature-channel dimensions widen the student-teacher gap that CWD's 1×1 projection head must bridge. SegNeXt-B offers the best capacity-compatibility trade-off for the 5M-param MobileNetV3-Large student.
MobileNetV2 or MobileViT as the student backbone	MobileNetV2 has documented PTQ catastrophic-failure modes (Krishnamoorthi, 2018). MobileViT lacks mature INT8 backend support and adds attention operators that quantize poorly. MobileNetV3-Large using its native Hard-Swish/ReLU activations through the

	quantizable TorchVision implementation is adopted as the lightweight student backbone. The global ReLU6 substitution is rejected because the reference quantizable implementation already supports MobileNetV3's native activations, while replacing Hard-Swish with ReLU6 risks avoidable FP32 accuracy loss without a clear INT8 deployment benefit.
DIST (Huang et al., 2022) with Pearson-correlation matching	DIST's inter-class relational structure becomes unreliable under PlantSeg's 115-class long-tail imbalance (many classes <20 training images per Wei et al., 2026). Logit KD + CWD adopted as more stable under severe class imbalance. DIST is retained as a fallback recovery method only if Logit KD fails to produce E2 improvement.
KD-assisted INT8 QAT (reapplying logit and channel-wise distillation during quantization-aware training)	Reapplying distillation during INT8 QAT would conflate the effect of quantization with continued teacher guidance and obscure the E3-versus-E6 and E5-versus-E6 accuracy-retention interpretation; supervised-only QAT is adopted to keep quantization-effect attribution clean. Evidence that distillation during low-precision training recovers accuracy is strongest at sub-8-bit precision and for compact depthwise-separable backbones (Mishra & Marr, 2018; Kim et al., 2019; Zhao & Zhao, 2024); KD-assisted QAT is therefore retained as the pre-registered E6-KD contingency, run only if the E3 → E6 mIoU drop exceeds 1.0 percentage point.
RobustBench (Croce et al., 2021) or Places2-C / NDS (Taori 2020; Kim 2021)	RobustBench targets adversarial robustness on classification; Places2-C and NDS benchmarks are not designed for dense prediction. Hendrycks–Dietterich + Kamann–Rother adopted as the appropriate framework for segmentation-task synthetic corruption robustness.

D. Research Instruments

1. Hardware and software instruments

The research instruments to be selected in this study are grouped into four functionalities: the model's learning and training (Python, PyTorch, MMSegmentation), applying corruptions and preprocessing (OpenCV for preprocessing; imagecorruptions library for synthetic corruption generation), testing robustness and efficiency (PyTorch profiling tools, `fvcore` (FlopCountAnalysis) for FLOPs profiling, and CPU proxy latency measurement), and evaluation and visualization (segmentation prediction overlays, pixel-level error maps, and class-confusion overlays). These were chosen as standalone tools appropriate for a given stage of the process, rather than as general-purpose instruments, to accommodate the need for FP32 training, teacher-student distillation, INT8 quantization, a corrupted test set, a CPU proxy benchmarking suite, and an interpretable visualization of features, respectively. Tool-specific preconditions (e.g., QNNPACK backend operator support for INT8 Sigmoid in the LR-ASPP head, MMSegmentation model-zoo checkpoint availability for SegNeXt-B, and RunPod GPU pod availability) are verified during pilot runs before E1 training begins. Any operator unsupported by the target INT8 backend is dequantized to FP32 (mixed-precision fallback) and the impact reported in the efficiency metrics. The eager-mode `torch.ao.quantization` workflow with the QNNPACK backend is selected for compatibility with the version-pinned environment required to fine-tune the SegNeXt-B teacher under MMSegmentation, which constrains the framework to a PyTorch release that fully supports eager-mode quantization. Graph-based alternatives, such as FX-mode and PyTorch-2-Export quantization, require a newer framework version that is incompatible with this pinned stack, and the choice of quantization front-end does not alter the underlying INT8 arithmetic being evaluated. Because the eager-mode workflow inserts observers and fuses operators explicitly, backend operator support is treated as a gating precondition rather than an assumption: a pilot quantization smoke test confirms that all

required operators, including the INT8 Sigmoid in the LR-ASPP global-pooling branch, are supported before the quantization stages are run.

Training will be conducted using RunPod as the primary cloud GPU platform. The selected RunPod GPU pod type, GPU model, VRAM capacity, CUDA image, storage configuration, and runtime duration of each training run will be documented. If multiple GPU pod types are used due to availability or budget constraints, the exact hardware used for each stage is reported. The cumulative GPU-hours and associated compute cost are logged per experimental stage, including the three-seed repetitions of E1 and E3, the per-seed quantization-aware fine-tunes, and the single teacher fine-tune, and reported in Chapter 4 so that total compute consumption and budget feasibility are transparent. In addition to wall-clock GPU-hours, the per-iteration training cost of each distillation stage is reported, separating the shared frozen-teacher forward pass — incurred by both E2 and E3 — from the marginal cost of the channel-wise distillation loss and its training-only projection head in E3, so that the compute difference between E2 and E3 can be expressed as a fraction of total training cost. The teacher forward-pass cost is profiled once using the same fvcore-based convention applied to the student efficiency metrics. CPU proxy benchmarking will be performed on a fixed CPU environment associated with the selected RunPod pod or a separately documented CPU instance, using single-threaded execution at batch size 1. The CPU model, thread count, RAM, and runtime configuration are reported together with the latency results. The software stack will be documented through a version-pinned environment manifest, hashed requirements lockfile, installation recipe, and Dockerfile or container image used on RunPod. The stack includes Python 3.11, PyTorch 2.1.0+cu121, MMSegmentation 1.2.2, mmcv 2.1.0, TorchVision 0.16.0, OpenCV 4.8.1, SciPy 1.11.4, NumPy 1.26.4, fvcore

0.1.5.post20221221, statsmodels, and vendored imagecorruptions functions rather than an unmodified imagecorruptions package dependency. The final confirmed RunPod GPU pod type, CPU model, CUDA/container image, storage configuration, and hardware configuration will be reported in Chapter 4.

Table 3.4 Computational Tools, Frameworks, and Model Architectures Used in the Study

Instrument	Role	Extent of Use
Python 3.11	Primary programming language	Entire experimental pipeline
PyTorch (Paszke et al., 2019)	Model definition, training, knowledge distillation, and quantization workflows	All seven student stages (E1–E7); teacher loaded from MMSegmentation model zoo
MMSegmentation	Segmentation framework for SegNeXt-B teacher model setup and training	Teacher preparation stage only. MobileNetV3-based student training is implemented directly in PyTorch using TorchVision's pretrained MobileNetV3-Large backbone, subject to architecture compatibility verification.
OpenCV (Bradski, 2000)	Image preprocessing (decoding, resizing, padding). Corruption application is handled by the imagecorruptions library (separate row). OpenCV is the underlying image I/O layer for both.	Preprocessing and test-set corruption generation
Albumentations	Joint image + mask augmentation	Applies identical geometric transforms to images and masks with per-target interpolation, using bilinear

		interpolation for images, nearest-neighbor interpolation for masks, and 255 fill for mask regions introduced by rotation or padding; version pinned in the reproducibility manifest.
RunPod	Cloud-based GPU compute environment	Model training and experimentation on a selected RunPod GPU pod. The exact GPU model, VRAM, CUDA/container image, disk configuration, run duration, and cost-relevant pod setting are documented for each run. Any pod interruption, hardware substitution, or migration between pod types is reported as a reproducibility.
Pinned environment lockfile and Dockerfile	Environment and dependency management	A version-pinned environment manifest, hashed requirements lockfile, installation recipe, and Dockerfile are committed to the project repository. The environment pins PyTorch 2.1.0, TorchVision 0.16.0, NumPy 1.26.4, SciPy 1.11.4, MMSegmentation 1.2.2 and mmev 2.1.0 (compatible CUDA builds), and all evaluation dependencies. Anaconda or venv alone is not treated as sufficient because it does not fully capture compiled

		CUDA/mmcv artifacts or cloud runtime variation.
PyTorch CPU inference benchmarking (torch.utils.benchmark)	CPU-proxy latency and peak memory benchmarking	All efficiency benchmarking is performed using PyTorch's CPU inference mode at batch size 1. Latency is measured with 20 warm-up runs discarded and 100 timed runs recorded, reporting median, interquartile range, and 95th-percentile latency. All timing runs use model.eval() and torch.inference_mode(), with automatic mixed precision disabled, CPU thread count fixed and reported, and all stages measured back-to-back in a single session on identical hardware. Results are labeled CPU proxy benchmarks. No ARM or embedded device testing is conducted.
Prediction-error maps and overlays (NumPy/Matplotlib)	Primary qualitative analysis	Per-pixel true-positive, false-positive, and false-negative overlays against the ground-truth mask, plus prediction panels and class-confusion overlays. These are produced for the critical stages including Teacher, E1, E3, and the converted INT8 E6, and may be shown for E2, E4, E5, and E7 in appendix panels where useful. Computed directly from model outputs without

		gradient flow, following the qualitative overlay approach of Kamann and Rother (2020).
SegNeXt (Guo et al., 2022)	Teacher model; upper-bound reference	Teacher training and benchmark comparison
MobileNetV3-Large with native Hard-Swish/ReLU activations and task-adapted LR-ASPP head (Howard et al., 2019; Jacob et al., 2018)	Lightweight student architecture	Student baseline (E1) and all student ablation variants (E2 through E7). MobileNetV3's native activations, including Hard-Swish and ReLU, are retained through the reference quantizable TorchVision implementation. The previously planned global ReLU6 substitution is not applied because the quantizable implementation already supports INT8 quantization of the native activation pattern, and replacing Hard-Swish would introduce an avoidable architecture change that may reduce FP32 accuracy without improving the selected INT8 workflow (Howard et al., 2019; Krishnamoorthi, 2018; Jacob et al., 2018). The LR-ASPP head is task-adapted with 256-channel intermediate classifiers sized to the empirically confirmed output-class count of the PlantSeg label space (Wei et al., 2026).

imagecorruptions (Hendrycks & Dietterich, 2019)	Synthetic image corruption generation for robustness evaluation	Applied at test time to E1 through E7 checkpoints and to the SegNeXt-B teacher reference. Generates motion blur, Gaussian noise, JPEG compression, brightness variation, and fog at severity levels 1 through 3 for summary and inferential metrics, with severity level 4 added for the descriptive degradation profile, on uint8 RGB test images before normalization. Severity 5 is excluded, and severity level 4 is retained for descriptive degradation profiling only (excluded from mIoU-C, RPD, rCD, and all inferential testing) because Kamann & Rother (2020) report severity-5 Gaussian noise on Cityscapes yielding SNR $\approx$ 1.7 dB, beyond any plausible field-acquisition envelope.
fvcore	FLOPs/MACs profiling	Used to profile the theoretical MACs of the student model at 512×512 input resolution, reported as MACs and as FLOPs = $2 \times$ MACs. Reported as a descriptive efficiency metric. FLOPs are not reduced by QAT and are profiled from the FP32 architecture.
SciPy (scipy.stats)	Statistical testing and analysis	Used for the Wilcoxon signed-rank test as primary (scipy.stats.wilcoxon with zero_method='pratt', alternative='greater',

		method='approx') and the paired t-test as a sensitivity check (scipy.stats.ttest_rel, alternative='greater'). Holm-Bonferroni correction is handled separately through statsmodels.
PyTorch torch.ao.quantization (QNNPACK backend)	INT8 PTQ and QAT implementation	Used for all quantization stages (E4, E5, E6, E7). Conv-BN-ReLU sequences are fused using torch.ao.quantization.fuse_modules() before quantization observer insertion; Hard-Swish and Hardsigmoid are quantized as standalone operators. Per-channel symmetric INT8 quantization is applied to all convolutional weight layers, while activations use asymmetric per-tensor UINT8 quantization with the full 8-bit activation range retained for the QNNPACK/ARM backend (Krishnamoorthi, 2018). Full INT8 inference is reported only after backend conversion and operator-support verification. (Jacob et al., 2018; Krishnamoorthi, 2018). QNNPACK backend operator support is verified during pilot quantization, including INT8 Sigmoid in the LR-ASPP global-pool branch (Howard et al., 2019). If a required operator is unsupported by the target backend, it is dequantized to

		FP32 (mixed-precision fallback) and reported in the efficiency metrics. Per-tensor asymmetric quantization is applied to activations; per-channel symmetric quantization is applied to all convolutional weight layers (Krishnamoorthi, 2018; Jacob et al., 2018).
TorchVision	Pretrained MobileNetV3-Large backbone initialization (TorchVision 0.16.0, MobileNet_V3_Large_Weights.IMAGENET1K_V2 checkpoint)	The native MobileNetV3 activations (Hard-Swish and ReLU) are retained through the quantizable TorchVision implementation; no activation substitution is applied to the loaded weights (Howard et al., 2019; Jacob et al., 2018). Used as the starting checkpoint for E1, which then becomes the starting point for E2 and E3.
PyTorch/NumPy/Python RNG seeding	Reproducibility	All training runs use random seed 42 across torch, numpy, and python's random module. <code>torch.backends.cudnn.deterministic = True,</code> <code>torch.backends.cudnn.benchmark = False,</code> <code>torch.use_deterministic_algorithms(True,</code> <code>warn_only=True),</code> and <code>CUBLAS_WORKSPACE_CONFIG=:4096:8</code> are set before CUDA initialization where applicable. Reproducibility is reported with the caveat that small

		floating-point variation may remain across GPU classes and compiled CUDA kernels. Three-seed validation is planned for E1 and E3; E4 and E7 are recomputed per seed from their corresponding FP32 checkpoints, while E5 and E6 are fine-tuned per seed where compute permits. Primary inference uses one pre-registered seed, with mean $\pm$ SD reported across completed seeds.
statsmodels	Multiple-comparison correction	Holm-Bonferroni correction across the 8 planned inferential tests via statsmodels.stats.multitest.multipletests(method='holm'). Returns family-wise corrected p-values used for hypothesis decisions.
scipy.stats.bootstrap	BCa bootstrap confidence intervals	Computes 95% BCa intervals (method='BCa', n_resamples=10000, confidence_level=0.95, fixed random seed) around per-image mean, median, and Hodges-Lehmann estimates and, through a paired custom statistic, the dataset-level pooled-mIoU difference. Where the BCa acceleration is degenerate, such as for a median on zero-inflated differences, a percentile or basic bootstrap interval is used and the method is reported.

2. Sources of data

Table 3.5 Data Sources, Descriptions, and Roles in the Study

Source Type	Specific Source	Role
Primary benchmark dataset	PlantSeg (Wei et al., 2026), Zenodo DOI: 10.5281/zenodo.17719108, CC BY-NC 4.0 license.	Training, validation, and clean/corrupted test evaluation
Derived robustness evaluation data	Corrupted version of the PlantSeg test set	Self-generated corrupted test set: five corruption types $\times$ severity levels 1–3 = 15 corrupted variants of each test image for summary and inferential metrics, plus severity level 4 for descriptive degradation profiling only
Pre-trained model weights	(1) MobileNetV3-Large: TorchVision MobileNet_V3_Large_Weights.IMAGENET1K_V2 (ImageNet-1K, top-1 accuracy 75.27%); used as the student backbone initialization for E1. (2) SegNeXt-B / MSCAN-B: MMSegmentation model zoo checkpoint segnext_mscan-b_512x512_160k_ade20k (or equivalent ADE20K-pretrained variant), fine-tuned on PlantSeg as the teacher.	Backbone initialization for SegNeXt and MobileNetV3-based models, subject to availability and compatibility
Self-generated experimental outputs	Training logs (TensorBoard or weights-and-biases), per-image and dataset-level metric CSVs, exported INT8 models (TorchScript	Primary empirical evidence for hypothesis testing and comparative analysis.

	.pt), checkpoint files (best-mIoU on val for each E_k), efficiency benchmark JSONs (latency, memory, FLOPs).	
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The proposed paper will exclusively utilize the PlantSeg (Wei et al., 2026) dataset for experimental purposes due to three distinct reasons. First, PlantSeg offers pixel-level class-index segmentation masks, accurately categorized across verified plant diseases, which is a prerequisite for both training and evaluating a semantic segmentation pipeline. Second, its image set exhibits natural in-the-wild characteristics, including variations in lighting, background clutter, and small lesion areas. These features provide a more representative benchmark to determine if a compressed student model can retain valuable segmentation capabilities after distillation and quantization, in contrast to controlled laboratory datasets such as PlantVillage (Mohanty et al., 2016) and Plant Pathology 2020 (Thapa et al., 2020), both of which provide image-level disease labels but not pixel-level lesion masks at the species/disease combinations PlantSeg covers. Third, the benchmark table presented by Wei et al. (2026) lists the published mean Intersection over Union (mIoU) and mean Accuracy (mAcc) metrics for the selected teacher model (SegNeXt-B / MSCAN-B) and the reference student architecture (MobileNetV3), offering essential performance reference points for interpreting the study's E1 baseline results. All results presented in this study pertain solely to PlantSeg and should not be extrapolated to other agricultural datasets, crop types, or imaging conditions without further evaluation (Wei et al., 2026; Pineau et al., 2021).

3. People involved in the study

This research is conducted by the four authors: Agalabia, Patrick Joseph C.; De Luna, Ainsley P.; Del Mundo, John Vincent A.; and Reyes, Zachary Martin D., under the supervision of Asst. Prof. Cherry Rose R. Estabillo as thesis adviser. The authors will be

responsible for the data preparation, model implementation, experiment execution, result output recording, and analysis. The thesis adviser will assist the authors with methodological issues and general supervision. No respondents are involved in this study. External plant pathology experts may be consulted only for qualitative interpretation of segmentation outputs. Their input does not alter ground-truth annotations, training data, model architecture, or experimental procedures, and is acknowledged but not authored on results. No ground-truth annotations will be generated by external experts for this study.

E. Sampling and Data Gathering Procedure

1. Sampling technique

This is a computational benchmarking study; no human respondents are sampled. The sampling units are annotated image-mask pairs drawn from the PlantSeg benchmark dataset (Wei et al., 2026). The samples are regarded as annotated image-mask pairs taken from the benchmark dataset, while outputs produced by the model, metrics, corrupted test images, and visual graphics are treated as experimental outputs rather than sampled entities. The available population of the study is the collection of appropriate plant disease imagery along with corresponding lesion masks present within the PlantSeg dataset itself and its equivalent evaluation subsets based on its official protocol (Wei et al., 2026).

Among the common sampling methods found in research methodology, purposive benchmark-dataset sampling is the appropriate technique. Probability-based methods (simple random, cluster, stratified, systematic random) assume the researcher draws a subset from a larger population, which does not apply here because the PlantSeg benchmark dataset is used in its entirety per its official splits. The reasoning behind this is that the research focuses on a fixed, publicly shared benchmark dataset that, in its structure, provides the necessary pixel-level ground-truth annotations for carrying out research related to semantic

segmentation. The use of PlantSeg is fitting for a benchmark comparison study since the dataset is reproducible and avoids biases associated with researcher-defined respondent selection (Weber et al., 2019; Wei et al., 2026).

This research will follow the established partition sets and their evaluation protocols provided by the dataset creators. This study uses the official PlantSeg dataset partition files: `annotation_train.json`, `annotation_val.json`, and `annotation_test.json`, which correspond to an approximate 70/10/20 train/validation/test split, yielding roughly 5,442 / 778 / 1,554 images respectively (computed from PlantSeg's 7,774-image total per Wei et al., 2026). A zero-overlap integrity check between partition identifier sets is performed before any training begins. The same split is used identically across all seven experimental stages. Since these datasets remain constant throughout the seven experimental stages (E1 to E7), only the model configuration and compression method are varied across the research experiments or ablation stages. Preprocessing steps, input resolution (512×512), normalization parameters, mask handling, evaluation metrics, and dataset splitting remain unchanged in all stages, ensuring that any observed differences in performance can be directly attributed to the compression method employed rather than variations in data management (Kerssies et al., 2024).

There is no separately curated dataset used as a dataset of corruption. Instead, the corruptions are generated on top of clean test images according to Hendrycks and Dietterich (2019) and Kamann and Rother (2020) semantic segmentation robustness research. Such corruptions are implemented to simulate common synthetic image degradations encountered during image acquisition, including motion blur, fog, and various forms of sensor or camera noise. These are considered controlled synthetic test-time stress conditions and are not intended to represent verified agricultural field environments. For the Post-Training Quantization stages (E4 and E7), a dedicated calibration subset of 128 images is randomly

sampled using seed 42 from the training partition, with the exact image identifiers persisted as a calibration-index file. This subset is used exclusively for the computation of activation statistics needed for quantization parameter estimation. A histogram observer is used for activation statistics; per-channel min/max observers for weights (Krishnamoorthi, 2018). The calibration images are preprocessed following the same deterministic pipeline as the clean test set, without applying any training augmentations. The subset is sampled to be representative of the natural class distribution under a fixed random seed rather than balanced across classes, since calibration estimates activation ranges and should match the inference-time distribution. Activation ranges are estimated with a histogram observer that selects the quantization range by minimizing quantization error, which clips outlier activations more robustly than a plain min/max estimator (Krishnamoorthi, 2018), while weight quantization parameters are computed per channel directly from the trained weights. The calibration configuration is selected on the validation partition, and final PTQ results are reported on the held-out test partition. The calibration subset does not participate in model weight updates and does not include any images from the validation or test partitions.

2. Data gathering procedure

The data will be collected by performing the following sequence of activities.

- a. **Dataset acquisition.** Obtain the PlantSeg images and corresponding lesion masks from the Zenodo repository (DOI: 10.5281/zenodo.17719108) (Wei et al., 2026). Before initiating any model training, verification proceeds per the checklist in Section C.1 (Table 3.1). Files failing any check are documented, removed, and excluded before training.
- b. **Data preprocessing.** Images and corresponding masks collected for the experiments in this paper are prepared following a strict protocol. These procedures are identical

across the official PlantSeg training set, the validation set and the test set to ensure the highest level of consistency and comparability in the outcomes of every phase of the study (Wei et al., 2026; Kerssies et al., 2024). Following verification (Section C.1, Table 3.1), images and masks undergo the preprocessing pipeline below, applied identically across the official PlantSeg training, validation, and test partitions. After successfully passing all verification checks, the remaining images and masks undergo aspect-ratio-preserving resizing, wherein the longer dimension is scaled to 512 pixels while retaining the aspect ratio. The shorter dimension is then symmetrically padded to reach 512 pixels: RGB images are padded with the per-channel ImageNet mean value ([124, 116, 104] in 8-bit terms), which maps to approximately zero after normalization. Mask annotations are padded with the integer value 255, which serves as the ignore label, excluding it from all loss calculations and evaluation metrics by setting `ignore_index=255`. The final spatial resolution of all images and masks after this padding process is 512×512 . For the images, bilinear interpolation is employed to preserve visual continuity, whereas nearest-neighbor interpolation is used for masks to accurately retain discrete class label boundaries. We then ensure all images are in the RGB format and scale pixel values to the $[0, 1]$ range. Subsequently, the per-channel normalization is done according to ImageNet statistics (mean = [0.485, 0.456, 0.406], standard deviation = [0.229, 0.224, 0.225]), in line with the ImageNet-pretrained weights of SegNeXt and MobileNetV3 (Guo et al., 2022; Howard et al., 2019). Finally, images will be cast into float32 tensors in the CHW [C, H, W] data order and the masks will be cast into int64 tensors in [H, W] format, with each tensor storing integer class indices identified from the PlantSeg grayscale annotations. The integer class indices in the mask tensors follow the class label mapping confirmed from the downloaded annotation files. Padded regions of the masks carry the value 255, which

is designated as the ignore label and are excluded from cross-entropy loss, Dice loss, and all segmentation metric calculations during training, validation, and evaluation. Unlike training augmentations, which are applied after padding, and test-time corruptions, which are applied after resizing but before padding, both preceding tensor conversion and normalization, this core preprocessing routine of aspect-ratio-preserving resize, padding, and ImageNet normalization is applied uniformly to every official partition of PlantSeg. The 512×512 input resolution is adopted to maintain comparability with the PlantSeg benchmark and the teacher reference. Wei et al. (2026) train and report their benchmark configurations at a 512×512 crop, and the SegNeXt-B teacher is fine-tuned at the same crop, so matching this resolution keeps the teacher features, the student inputs, and the published reference points on a common spatial footing. The resolution also represents a practical balance for the task: it is large enough to preserve the fine spatial detail required to delineate small lesion regions, yet small enough to keep training and CPU-proxy benchmarking feasible under the study's compute constraints. The same resolution is applied identically across all seven stages so that observed differences are attributable to the compression interventions rather than to input-scale variation.

- c. **Training-Time Augmentation.** Following preprocessing, the proponents will selectively apply augmentation to the training subset of the PlantSeg dataset. However, the validation and test sets (including both clean and corrupted) remain unaltered. This choice is made to guarantee a fair and replicable testing environment (Kerssies et al., 2024). The chosen augmentation strategy encompasses geometric transformations applied uniformly to both the image and its mask: random horizontal flip (probability 0.5) and random vertical flip (probability 0.5); random rotation of $\pm 10^\circ$ (probability 0.5), applied before cropping, where image regions newly created

by rotation are filled with the per-channel ImageNet mean and the corresponding mask regions are filled with the ignore label value 255; and multi-scale random resized cropping to a final resolution of 512×512 , with the resize scale drawn from the range $[0.75, 2.0]$ under aspect-ratio preservation and a single-class crop-rejection threshold (`cat_max_ratio`) of 0.95. For mask resizing during the crop operation, nearest-neighbor interpolation is used to preserve the integrity of the integer class labels. In addition to geometric augmentations, the approach includes an image-only photometric transformation: random color jittering restricted to hue (± 0.015) and saturation (factor within $[0.8, 1.2]$), applied with probability 0.5. Brightness and contrast jittering are deliberately excluded, both because leaf and lesion color is a primary diagnostic feature that aggressive photometric shifts would distort and because brightness and contrast overlap the brightness and fog corruptions used in the robustness evaluation. The corresponding masks are left unchanged to maintain the spatial relationship to the class labels. Blurring and noise augmentations are deliberately omitted from the training augmentation set. This decision is made because Gaussian noise, motion blur, and JPEG compression are among the five corruption types employed in the robustness evaluation stage (Hendrycks & Dietterich, 2019; Kamann & Rother, 2020). Including them during training would prematurely expose the model to conditions against which it is being assessed, thus undermining the comparative analysis of robustness across E1, E3, E4, E5, E6, and E7. This corresponds to the data-contamination prohibition in Hendrycks & Dietterich (2019): networks must not be trained on the corruption types used for evaluation. Class-aware weighting will also be adopted during training to reduce the class disparity among various diseases represented in the PlantSeg dataset.

Corruption constitutes a third, independent procedure that operates separately from both preprocessing and training augmentations. The precise methodology for obtaining the corrupted dataset involves the following steps: clean PlantSeg test images are selected, followed by an aspect-ratio-preserving resize. The chosen corruption is then applied to the resized uint8 RGB image, after which mean-value padding (image: per-channel ImageNet mean $\approx [124, 116, 104]$; mask: 255) is applied prior to tensor conversion and normalization. Next, the image is converted to a float tensor and undergoes ImageNet normalization. This corrupted normalized image tensor is then used in conjunction with the unmodified mask tensor for the robustness evaluation. Because these corruption functions include stochastic components, the NumPy random seed is fixed during corruption generation, and the resulting corrupted image set is cached and checksummed. All evaluated stages are therefore scored on byte-identical corrupted images rather than regenerating corruptions separately per model. Applying corruptions before normalization is essential because the vendored corruption functions based on Hendrycks and Dietterich (2019) operate on uint8 pixel values within the range $[0, 255]$. If corruption were applied post-normalization, it would not accurately reflect any plausible image acquisition degradation process. Padding is applied only after corruption so that spatially extended corruptions, such as motion blur and fog, act exclusively on genuine image content and do not blend the padded border into scored pixels near the image edges. The corrupted test images are generated using five corruption functions vendored from the imagecorruptions reference implementation (Hendrycks & Dietterich, 2019), rather than relying on the installed package, with deprecated NumPy aliases patched where needed for reproducibility. The five corruption types are motion blur, Gaussian noise, JPEG compression, brightness variation, and fog. For summary and inferential robustness metrics, these five corruptions are applied at severity levels 1, 2, and 3. The five corruption types are selected as a focused subset of the broader corruption suite of Hendrycks and Dietterich

(2019) on the basis of acquisition realism for handheld and in-field plant photography. They correspond, respectively, to camera or subject movement, sensor noise under low-light capture, lossy storage and transmission, illumination change, and atmospheric scattering, all of which are plausible during ordinary image acquisition. Corruption families that are characteristic of other domains or less representative of this acquisition setting, such as snow, frost, and elastic or pixelation transforms, are excluded so that the robustness evaluation remains interpretable for the target imaging context. Restricting the evaluation to a defensible subset rather than the full suite follows the segmentation-robustness ablation practice of Kamann and Rother (2020).

To additionally characterize degradation trajectory, each corruption is also evaluated at severity level 4 for descriptive per-corruption degradation profiling only. Severity level 5 is excluded because it corresponds to an unrealistically extreme degradation condition; severity level 4 is retained only for descriptive degradation profiling and is excluded from mIoU-C, RPD, rCD, and all inferential testing. Severity 1–3 represents the moderate-to-strong band Kamann & Rother (2020) used for segmentation ablation analysis. This methodology aligns with the approach presented in Hendrycks and Dietterich (2019) and adapted for assessing segmentation robustness in Kamann and Rother (2020). Unlike augmentations, the motivation here is not to boost the dataset diversity for training purposes but to conduct stress-testing of the models against degraded imagery at inference time in a controlled environment. To promote consistent findings and ensure comparability among all experimented model candidates, an unvarying corruption type and intensity range will be maintained for all evaluations (Keressies et al., 2024).

- d. **Training, validation, and ablation output collection.** The training data is used for training and tuning the models while the clean test data helps in evaluating final

model effectiveness. Clean test set images are passed through all seven experimental stage checkpoints (E1 through E7), as well as the SegNeXt-B teacher model reference, to generate prediction maps and per-image segmentation outputs. Dataset-level mIoU, Dice Coefficient, and mAcc are calculated for each experimental stage. Per-image mIoU is recorded for all 1,554 test images and all stages (E1–E7 + teacher), stored as a CSV with columns {image_id, stage, mIoU, Dice, mAcc}. These per-image values are the input to all 8 paired inferential tests in Section F.1.b. The generated prediction maps, summary metric tables, and per-image metric arrays are stored in files that are clearly labeled according to their corresponding experimental stage. Prediction maps are stored as PNG files in `outputs/<stage>/predictions/`; summary score tables as CSVs in `outputs/<stage>/metrics/`; runtime measurements as JSON in `outputs/<stage>/efficiency/`. Filenames encode the experimental stage and (where applicable) corruption type and severity. For each FP32 stage (E1–E3), the checkpoint with the highest mIoU on the validation partition is retained as that stage's evaluated checkpoint. For INT8 stages (E4–E7), the final post-quantization checkpoint is the evaluated checkpoint (no validation-mIoU selection because INT8 inference is deterministic post-conversion).

- e. **Corruption-based robustness evaluation data gathering.** Corruptions specified in E.2.4 are applied to E1 through E7 and, as a descriptive reference, to the SegNeXt-B teacher. E2 is included to decompose the distillation contribution to robustness, while the teacher provides an upper bound on absolute corrupted performance and a retention reference. For each (stage $\times$ corruption $\times$ severity) cell, per-image mIoU-C is recorded for the 1,554 test images. $RPD = (mIoU_clean - mIoU_C) / mIoU_clean \times 100$ is computed per stage and corruption category. rCD is computed descriptively

for E1 through E7 using E1 as the internal reference; the teacher is excluded from rCD. Results are stored as CSVs in outputs/<stage>/robustness/.

- f. **Efficiency benchmarking data gathering.** Model size, parameter count, FLOP count, CPU-runtime latency, and peak memory usage will be gathered. Efficiency metrics are predominantly collected for the student model configurations (E1 through E7). For the SegNeXt-B teacher model, model size and parameter count are provided solely as descriptive context to illustrate the extent of compression achieved by the student model. The teacher model is not intended for deployment and its latency and memory measurements are not treated as relevant benchmarks for deployment purposes.
- g. **Qualitative data gathering through prediction overlays and error maps.** Pixel-level error maps and prediction-overlay panels are produced for the Teacher, E1, E3, and E6, with selected appendix panels for E2, E4, E5, and E7 where useful. Error maps compare each prediction against the corresponding ground-truth mask by highlighting true positives, false positives, and false negatives, following the qualitative overlay approach used in semantic-segmentation robustness analysis by Kamann and Rother (2020). Class-confusion overlays are also generated to show which disease classes are confused with one another in the 115-class disease label space. These visualizations are computed directly from model outputs and therefore apply to both FP32 checkpoints and the converted INT8 E6 model without requiring gradient flow or surrogate visualization.

The outputs collected from the same held-out PlantSeg test images will be used to build composite figures. Representative visual plates will be presented, showcasing the initial input image, its corresponding ground truth mask, and the segmentation outcomes for the Teacher, E1, E3, and E6 on representative images from the PlantSeg

test set. These four configurations are chosen to represent the key progression: the upper-bound reference (Teacher), the undistilled FP32 baseline (E1), the distilled FP32 student model (E3), and the proposed compressed pipeline (E6). The remaining experimental stages (E2, E4, E5, E7) will be reported quantitatively in the results tables; selected qualitative panels for these stages may be included in the appendix, facilitating direct comparison of differences in lesion boundary characteristics, coverage rates, and class-region alignment. Similarly, visual comparisons between the clean and corrupted conditions of the same held-out PlantSeg test images will be generated. The corrupted test conditions will sample three of the five selected corruption types: motion blur, Gaussian noise, and fog. Each corruption will be applied at severity level 2, which represents the mid-range setting within the severity 1–3 protocol, with side-by-side visual outputs from E1, E3, and E6 accompanying them, to explicitly clarify the meanings of mIoU-C and RPD within the framework defined by Hendrycks and Dietterich (2019) and adapted by Kamann and Rother (2020) for segmentation. Pixel-level error maps and prediction-overlay panels are produced for the Teacher, E1, E3, and E6, with selected appendix panels for E2, E4, E5, and E7 where useful. Error maps compare each prediction against the corresponding ground-truth mask by highlighting true positives, false positives, and false negatives, following the qualitative overlay approach used in semantic-segmentation robustness analysis by Kamann and Rother (2020). Class-confusion overlays are also generated to show which disease classes are confused with one another in the 115-class disease label space. These visualizations are computed directly from model outputs and therefore apply to both FP32 checkpoints and the converted INT8 E6 model without requiring gradient flow or surrogate visualization. The entire visual component will originate from the same

held-out PlantSeg test images, promoting reproducible visual benchmarking throughout the paper. The generated visuals are reused across the thesis chapters, including Chapter 4 and Chapter 5.

F. Statistical Treatment of the Data

The present work adopts a descriptive, comparative, metric-based, and, where appropriate, inferential statistical treatment of data. Due to the focus of the work on benchmarking computer vision, rather than the analysis of human participants or qualitative feedback, the primary approach is the comparative performance evaluation of the seven student configurations (E1–E7) on per-image and dataset-level metrics. The SegNeXt-B teacher reference is reported descriptively only as an upper-bound (Table 3.6, Section A H.1d).

Descriptive statistics will be used to quantify the performance of all models considered, reporting average performance on the entire test set, as well as mean and standard deviation of model outputs if multiple training seeds are available. To characterize run-to-run variability, the two full-precision trainings that anchor the distillation comparison, E1 and E3, are each repeated under three random seeds; the post-training-quantized stages derived from them (E4 and E7) are recomputed per seed at no additional training cost, and the quantization-aware stages (E5 and E6) are fine-tuned once per seed. The SegNeXt-B teacher, a fixed upper-bound reference outside the inferential family, is trained once. For every multi-seed stage, the mean and standard deviation of dataset-level mIoU across seeds are reported. Primary inferential testing is conducted on a single pre-registered seed fixed in advance, and the remaining seeds confirm that the total distillation gain ($E1 \rightarrow E3$) and the quantization-induced drop ($E3 \rightarrow E6$) are stable across initializations and exceed the observed run-to-run variation. Repetition of the Logit-KD-only stage (E2) is treated as optional; if it is not completed, the decomposition of the distillation gain into its Logit-KD

and CWD components is reported as conditional on a single initialization. No frequency-distribution analysis is performed, as the study has no categorical-response variables of the kind such tests address. The class distribution of PlantSeg is reported descriptively to characterize the dataset's imbalance. Nevertheless, the class distribution within the dataset will be shown to illustrate imbalance in disease categories within the PlantSeg dataset. Linear correlation and regression are not applied, as the purpose of the study is not to relate different variables of model performance to predict future outcomes, but to compare models under precisely the same experimental constraints.

Apart from descriptive benchmarks, inferential hypothesis testing will be utilized where sufficient paired and repeatable observations can be provided for each comparison. In terms of segmentation accuracy, the mean Intersection over Union (mIoU) will be calculated at the dataset level as the primary reported metric, while per-image mIoU will be used as the paired observation unit for inferential testing. The Dice Coefficient is reported descriptively at dataset level as a secondary metric but is not subjected to separate inferential testing, because Dice and mIoU are closely related at the class level and would likely produce redundant test results while increasing the multiple comparison burden. Since the same set of test images will be used in the performance comparisons of all models, the data points in each pairwise comparison can be considered as paired observations (Nariya et al., 2023). The primary statistical test for the planned pairwise comparisons is the one-sided Wilcoxon signed-rank test on per-image differences in mIoU values (Wilcoxon, 1945; Pratt, 1959; Conover, 1973). The Wilcoxon signed-rank test is selected over the paired t-test because the data contains scores on the $[0, 1]$ range, is zero for images in which both models miss small lesions, and could be multimodal or skewed in shape (Wiesenfarth et al., 2021; Demšar, 2006). Primary result summaries rely on non-parametric measures that might be more descriptive for data fitting such criteria, compared to mean-based inferences like the t-test.

Still, the paired t-test using `scipy.stats.ttest_rel` is computed and reported in each case as a robustness check. If both tests agree, both are reported, with the Wilcoxon one preferred. If they disagree, both results are still reported, with the Wilcoxon conclusion preferred, and a discussion of why one might have gotten a positive result and the other negative. At $n = 1,554$, the Central Limit Theorem implies that paired t-tests should be reasonably accurate to be used as checks (Rainio et al., 2024). Since there are 1,554 per-image differences at hand, the Shapiro-Wilk test is not used for gatekeeping: at this sample size, it may well reject normality just due to tiny deviations that are practically irrelevant (Rainio et al., 2024; Shapiro & Wilk, 1965). Across the 8 planned tests below, a familywise significance level of $\alpha = 0.05$ is controlled by applying the Holm-Bonferroni correction, described in Section F.1.d.

For robustness evaluation, a similar paired-testing approach will be adopted for per-image corrupted mIoU, or per-image drops in accuracy, when the experimental design permits comparable paired data. For purely deterministic efficiency measures (physical file size of the model, total parameter count, FLOPs), hypothesis testing will not be used as these metrics directly stem from the network's architecture, the export procedure, or the profiling method and can't be treated as outcomes of an experiment. For dynamic performance-based efficiency measures (CPU inference time, peak memory consumption), the models will be run multiple times using identical hardware configurations and runtime conditions. Descriptive statistics, including mean, standard deviation, and where relevant, paired significance tests, will be presented. If any re-training runs were completed, they'd only be presented as supportive results demonstrating model consistency rather than forming the core component of any statistical comparisons, unless identically set up and performed multi-run experiments are available for all configurations tested.

The interpretation of results herein takes a form based on hierarchical decision-making rather than presenting all metrics on an even footing. The primary decision factor is whether E6 simultaneously satisfies four operational criteria (Section A, H1d): (i) clean-test mIoU(E6) within 2.0 pp of mIoU(E3); (ii) per-image mIoU(E6) > per-image mIoU(E1); (iii) per-image mIoU-C(E6) > per-image mIoU-C(E1); (iv) descriptive efficiency reductions relative to FP32 student variants. The planned inferential comparisons, each using the one-tailed Wilcoxon signed-rank test as primary and the paired t-test as sensitivity check, are: E1 vs E2 (effect of Logit KD); E2 vs E3 (added value of CWD beyond Logit KD); E4 vs E5 (PTQ vs QAT on the undistilled student); E7 vs E6 (PTQ vs QAT on the distilled student); E4 vs E7 (effect of distillation on the PTQ path); E5 vs E6 (effect of distillation on the QAT path); E1 vs E6 (full pipeline vs baseline); and one robustness inferential test, E1 vs E6, based on per-image mIoU-C. These total eight tests are subject to Holm-Bonferroni correction (Holm, 1979). Primary accuracy results for the clean test set, including mIoU and Dice Coefficient, are reported descriptively for all seven experiment stages (E1–E7) along with the SegNeXt-B teacher. The 2.0 percentage-point margin is grounded in the established INT8 accuracy-retention envelope for per-channel-quantized convolutional networks; for MobileNet-based segmentation models, per-channel INT8 quantization has been shown to retain mIoU within approximately one percentage point of full precision, so 2.0 percentage points is adopted as a deliberately conservative bound for the more challenging 115-class, low-mIoU regime examined in this study (Jacob et al., 2018; Krishnamoorthi, 2018; Nagel et al., 2019). It is noted that compact depthwise-separable backbones such as MobileNetV3-Large can retain a marginally larger residual gap at 8-bit precision than larger convolutional backbones, even under quantization-aware training, which further supports adopting a conservative retention bound and pre-registering the E6-KD contingency (Zhao & Zhao, 2024). E6 is judged non-inferior to E3 if the lower bound of the paired BCa 95%

confidence interval for the dataset-level mIoU difference, $\text{mIoU}(\text{E6}) - \text{mIoU}(\text{E3})$, remains above -2.0 percentage points. This non-inferiority check is reported separately from the eight-test Holm-Bonferroni superiority family. As a separate, stricter pre-registered trigger, the distillation-assisted contingency arm (E6-KD) is run if the observed $\text{E3} \rightarrow \text{E6}$ reduction in clean-test mIoU exceeds 1.0 percentage point, the empirical per-channel INT8 retention envelope reported for segmentation models (Nagel et al., 2019). This 1.0-point trigger for running the contingency is distinct from, and stricter than, the 2.0-point margin used for the formal E3-versus-E6 non-inferiority decision. The actual $\text{E3} \rightarrow \text{E6}$ drop, the relative retention of E3 mIoU, and sensitivity conclusions at margins of 1.0, 1.5, 2.0, and 2.5 percentage points are reported so the dependence of the conclusion on the selected margin is transparent. E1 serves solely as the raw baseline for gauging the overall end-to-end change throughout the process.

1. Inferential hypothesis testing

To test the hypotheses outlined in the study, an inferential testing framework will be employed to analyze the paired per-image metrics that arise from the planned model comparison. The study maintains a distinction between the dataset-level mIoU, which represents a single score by first computing the aggregate intersection and union over all test images per class, then taking a macro-average of IoU per class, and per-image mIoU, which measures improvement as the macro-average of per-image class IoU for those classes present in that image, providing a single observation per test image. Per-image mIoU serves as the basis for paired inferential testing due to one observation per test image, while dataset-level mIoU will be assessed directly by bootstrap confidence intervals separately. Per-image mIoU is the macro-average of class IoU values for classes with non-zero ground-truth pixels in that image (absent-class exclusion rule); classes with zero ground-truth pixels in a given image

are excluded from that image's denominator. Dice Coefficient will be described only at the dataset level; there will be no separate hypothesis tests against it, since it is strongly related to mIoU and thus provides limited added value (Müller et al. 2022, Ren et al. 2025). Mean accuracy (mAcc) is reported descriptively to benchmark against Wei et al. (2026). All reported efficiency metrics are descriptive only. CPU proxy latency, peak memory, number of parameters, model size and FLOPs are either objective properties of the models or measurements whose results rely heavily on hardware specificity and the conditions under which they are measured. On-device mobile latency is not claimed; establishing it would require lowering to an on-device runtime such as ExecuTorch with the XNNPACK delegate and measurement on representative ARM hardware, which is outside the scope of this study. The comparison of Teacher vs. E6 serves as a non-operative upper-bound illustration rather than a standard of performance.

a. Normality assessment

Because the Wilcoxon signed-rank test will be the main method for statistical inference in these tests, a formal assessment of the normality assumption of paired differences is not necessary. The analysis relies on the approximate assumption that the paired differences are at least approximately symmetric around the null shift. The paired Δ mIoU distributions on PlantSeg test images are heavily zero-inflated, particularly in comparisons involving stages with similar overall accuracy (e.g., E2 vs E3, E3 vs E6), where many test images yield identical per-image mIoU under both models. The paired t-test sensitivity check (F.1.b) addresses this differently: it relies on the Central Limit Theorem for the mean difference at $n \approx 1,554$, not on symmetry of the underlying distribution. In terms of practical application, the CLT effect might not fully address the normality concern. Shapiro-Wilk and similar normality tests are not used as gatekeepers because at $n \approx 1,554$

they may reject normality for deviations of practically negligible size (Rainio et al., 2024).

The paired differences are computed as:

$$d_i = A_i - B_i$$

with A_i and B_i referring to the per-image mIoU values for Models A and B across test images, respectively.

b. Pairwise comparison - Wilcoxon Signed-Rank Test

All of the comparisons rely on a paired Wilcoxon signed-rank test applied to per-image Δ mIoU differences. The per-image mIoU score resides on $[0, 1]$ and includes a 0 point that occurs frequently, specifically when the models completely fail to detect small lesions. Furthermore, per-image Δ mIoU has some positive skew when many test images show missed detections and is possibly bimodal, making mean improvement less representative. Also, paired tests on per-image segmentation metrics are an established practice in the medical and natural-image segmentation literature (Wiesenfarth et al., 2021; Maier-Hein et al., 2018, 2024). The paired t-test will be included as a pre-registered sensitivity analysis because it provides an interpretable estimate of mean improvement. This is useful for reporting engineering-relevant magnitude information that rank-based tests, such as the Wilcoxon signed-rank test, do not directly provide. In the cases below, only the Wilcoxon test results drive the primary conclusions, but both tests will be included together in the table.

The Wilcoxon signed-rank test is computed as `scipy.stats.wilcoxon(d, zero_method='pratt', alternative='greater')`. The method 'pratt' (Pratt 1959, Conover 1973) integrates the zero values with the nonzero values when determining ranks and omits their summation, a

necessary step given the situation presented by the PlantSeg paired ΔmIoU that are heavily zero-inflated. Since the paired difference d_i was created and described earlier and H_{1a-d} explicitly direct attention toward improvement rather than some deviation, the tests were performed one-tailed, all at the family-wise significance level $\alpha = 0.05$.

Since the sample size is large, the test statistic W for the Wilcoxon signed-rank test is calculated from its large-sample normal approximation. This creates a normalized z-statistic:

$$z = \frac{W - \mu_W}{\sigma_W}$$

with $\mu_W = \frac{n(n+1)}{4}$ and $\sigma_W = \sqrt{\frac{n(n+1)(2n+1)}{24}}$, where σ_W is reduced by the tie-correction term

$\Sigma(t_i^3 - t_i)/48$, with t_i denoting the size of each group of tied ranks (Conover, 1973), giving

$$\sigma_W = \sqrt{[n(n+1)(2n+1)/24 - \Sigma(t_i^3 - t_i)/48]};$$

a continuity correction of ± 0.5 is additionally applied to W before computing z . Under Pratt's method (Pratt, 1959), zero differences are assigned ranks before being excluded from R^+ and R^- , preserving their contribution to the variance rather than discarding them entirely. Both adjustments are applied via `scipy.stats.wilcoxon(d, zero_method='pratt', correction=True, alternative='greater')`. The W , z and Holm-adjusted p-value of each comparison pair are reported.

The sensitivity is analyzed next using a paired t-test statistic:

$$t_{calc} = \frac{\bar{d}}{s_d/\sqrt{n}}$$

where $\bar{d}$ and s_d denote the mean and standard deviation of the differences between paired observations ($n \approx 1,554$), respectively. If conclusions differ between tests, Wilcoxon will be preferred, with both tests reported, along with further context.

For all planned hypotheses tested, alongside W, z and the Holm-adjusted p-value, a collection of statistics characterizing effect size is reported. The matched-pairs rank-biserial correlation r_{rb} for the Wilcoxon tests with range $[-1, 1]$:

$$r_{rb} = \frac{R^+ - R^-}{\Sigma R}$$

with R^+ and R^- the rank sums of positive and negative differences. The Hodges-Lehmann pseudomedian shift estimates the median value of Walsh averages (Hodges & Lehmann 1963), which is the best median point estimate associated with Wilcoxon signed-rank tests. Also, d_z for the paired differences d :

$$d_z = \frac{\text{mean}(d)}{SD(d)}$$

(Cohen 1988) is computed with benchmarks of small (0.20), medium (0.50) and large (0.80) for convenient interpretation alongside paired t-test measures. Lastly, the mean percentage point ΔmIoU is reported as the engineering-interpretable mean-shift value, as improvement can only be considered if both statistical significance is reached, but not used as the sole criterion, and if the absolute shift is sizable and supported by the effect sizes.

c. Supplementary validation: Bootstrap confidence intervals

Through 10,000 bootstrap replicates, BCa bootstrap confidence intervals are calculated (Efron & Tibshirani 1993, Efron 2008) around both the dataset-level mIoU difference achieved through bootstrapping over test images and around the dataset-level mean, median and Hodges-Lehmann estimates computed on per-image scores for purposes of providing uncertainty estimates to the Wilcoxon signed-rank primary analysis and the t-test secondary analysis. The 95% interval reported is a 95% BCa interval (Efron & Tibshirani 1993), commonly considered a more reliable and less biased confidence interval calculation for large samples than simpler bootstrap interval methods, on data with positive skew like the segmentation accuracy results collected herein. By performing bootstrapping across the same test images used for evaluating the models' predictive performance, the chosen benchmark metric is directly corroborated in its evaluation at the full dataset level, not solely by its per-image evaluation. This dual-level validation matters because dataset-level mIoU is computed by aggregating intersections/unions before averaging, so dataset-level and per-image conclusions can diverge even when both are technically valid. If the BCa interval excludes zero in the predicted direction, the bootstrap supports the Wilcoxon conclusion. If the BCa interval includes zero while the Wilcoxon rejects H_0 , the result is treated as more weakly supported and the divergence is discussed.

d. Significance level and decision criteria

Eight planned statistical hypothesis tests are to be carried out, seven per-image mIoU differences and one per-image mIoU-C difference comparison of E1 versus E6, for which Holm-Bonferroni correction for multiple testing will be applied. To do this, each of the eight resulting p-values, labeled 1–8, is given a ranking starting with the lowest, rank 1, up to rank 8. The k th lowest value p_k is then evaluated for the adjusted threshold:

$$p < \frac{\alpha}{9-k}$$

where $\alpha = 0.05$. When this inequality holds, the comparison is judged statistically significant; otherwise, the null hypothesis is not rejected at this ranking. This must be achieved, alongside acceptable magnitude and effect size, before claiming superiority for a comparison. All 8 tests use the same $n \approx 1,554$ paired observations, so no within-test sample-size adjustment is needed.

2. Segmentation Accuracy Metrics: Primary and Secondary

Mean Intersection over Union (mIoU). The mean Intersection over Union (mIoU) is selected as the primary accuracy metric for three reasons dictated by the study design. First, Wei et al. (2026) reports it as the main metric for the PlantSeg benchmark, so mIoU allows contextual comparison between the E1 baseline and published reference values. Second, in comparison with Dice, mIoU may be more stringent in penalizing over-segmentation and under-segmentation, which is useful for a multiclass segmentation task where misclassifying a healthy region into one of the disease categories and skipping a disease lesion are both meaningful errors. Finally, mIoU serves as a standard evaluation metric in several common semantic segmentation benchmarks like ADE20K, Cityscapes, and Pascal VOC, which means this study remains aligned with general semantic segmentation reporting practice. Pixel accuracy is not used as the primary metric because it can be heavily biased by background pixels, which constitute a large fraction of many plant images and may lead to misleadingly high scores even if the model fails on lesions (Müller et al., 2022; Ren et al., 2025). Pixels labeled 255, corresponding to padded canvas regions, are ignored during IoU calculation. This avoids inflating accuracy scores from non-relevant padded regions. The mIoU value is calculated according to the following formula:

$$mIoU = \frac{1}{N} \sum_i^N \frac{TP_i}{TP_i + FP_i + FN_i}$$

Where N is the number of classes averaged, TP_i is the pixels correctly predicted as class i , FP_i is the background, healthy-leaf, or other-class pixels mistakenly assigned to class i , and FN_i is the class- i pixels not detected. Two variants are reported. The dataset-level mIoU accumulates TP, FP, and FN per class across the entire test set and averages over all 115 verified classes, consisting of the background class and 115 disease classes. This all-class dataset-level mIoU matches the PlantSeg benchmark reporting and serves as the basis for dataset-level non-inferiority and bootstrap analyses. The per-image mIoU used as the paired unit for Wilcoxon tests is computed over the 115 disease classes only, with the background class excluded to avoid diluting lesion-level sensitivity. For each image, disease classes with zero ground-truth pixels in that image are excluded from that image's average, so the averaged class set and its size are determined by the ground truth and are identical for every model compared on that image. Pixels labeled 255 are excluded from all computations. Because the per-image average omits disease classes absent from the ground truth, it does not penalize false-positive predictions of absent disease classes; that penalty is captured by dataset-level mIoU, where such predictions enter each class union. Dice and mAcc are reported in both disease-only and all-class form when useful, but inferential testing uses per-image disease-only mIoU.

Dice Coefficient (F1-Score). Dice Coefficient (F1-Score) is the secondary accuracy metric in the evaluation. It corresponds to the harmonic mean of precision and recall values, imposing penalties to false positives and false negatives equally. As with IoU, $Dice = 2 \cdot IoU / (1 + IoU)$, so Dice is a monotonic function of IoU at the class level. Consequently, for many common comparisons, models tend to retain similar ranking patterns in Dice and IoU

measures. Dice is chosen as a secondary measurement for two main reasons (Müller et al., 2022; Ren et al., 2025). First, as a constituent component of the combined CE + Dice objective loss for supervised training, it contributes optimization stability and robustness to class imbalance through its region-overlap normalization. Second, Dice is relevant to small lesion detection because PlantSeg is cited with high challenges in lesion detection, since about 80% of images have lesion masks below 36% (Wei et al., 2026). Due to its close relationship with IoU, Dice will only be provided as descriptive statistics and will not be subjected to repeated statistical tests. This measure is obtained by:

$$Dice = \frac{2 \times TP}{2 \times TP + FP + FN}$$

Reported only at the dataset level, per-image Dice is not computed because it would be redundant with per-image mIoU under the monotonic $Dice = 2 \cdot IoU / (1 + IoU)$ relationship.

Mean Accuracy (mAcc). Mean accuracy will only serve as a descriptive accuracy comparison tool. It is computed as average pixel accuracy for each verified class.

$$mAcc = 1 / |C| \sum_{c \in C} TP_c / (TP_c + FN_c)$$

where C denotes the set of all classes, TP_c is the number of correctly predicted pixels for class c , and FN_c is the number of ground-truth pixels mispredicted from class c . Because mAcc does not penalize false positives, it is generally insufficient on its own as a segmentation quality measure compared to mIoU or Dice (Müller et al., 2022; Ren et al., 2025). Nonetheless, it is valuable in this context because Wei et al. (2026) reports both mAcc and mIoU as benchmark figures for PlantSeg, allowing contextual comparison against their findings. The study excludes mAcc from further inferential analyses as part of segmentation metric comparison. Boundary quality is not adopted as a separate evaluation metric; segmentation accuracy is reported through mean IoU and the Dice

coefficient. Boundary IoU is an object-centric measure designed primarily to expose boundary errors on large objects that interior-dominated mask IoU may obscure, and it requires pixel-precise reference contours to constitute a valid model-quality signal. PlantSeg masks are polygon annotations with many lesions having fragmented boundaries and gradual color transitions into healthy tissue; under these conditions, a boundary-band metric may partly reflect annotation variability rather than model boundary fidelity. Plant lesions also frequently occupy small image regions, where mask IoU is already boundary-sensitive. Mean IoU and Dice therefore provide a benchmark-comparable overlap-based assessment for this task, while Boundary IoU is excluded from the primary and secondary metric set.

3. Robustness metrics

Corrupted mIoU (mIoU-C). In this paper, Corrupted mIoU (mIoU-C) is computed by applying the standard mIoU formula to the corrupted versions of the PlantSeg test set. The primary reported mIoU-C value is the mean of per-corruption-type mIoU scores, each averaged over severity levels 1, 2, and 3, across all five selected corruption types: motion blur, Gaussian noise, JPEG compression, brightness variation, and fog (Hendrycks & Dietterich, 2019; Kamann & Rother, 2020). This yields a single scalar robustness summary per model configuration. The same absent-class exclusion rule applied to clean mIoU is enforced: classes with zero ground-truth pixels in a given corrupted test image are excluded from that image's per-image IoU average. Padded pixels (ignore label 255) are excluded from all computations. mIoU-C is computed for E1 through E7 and for the SegNeXt-B teacher as a descriptive reference. E2 is included to decompose the distillation contribution to corrupted performance; the teacher is reported for mIoU-C and RPD only and is excluded from rCD. The one inferential robustness test uses per-image mIoU-C values from E1 and E6 in a one-tailed Wilcoxon signed-rank test (primary) with paired t-test as sensitivity check, subject

to the same Holm-Bonferroni correction applied across all 8 inferential tests (F.1.d). The standard mIoU formula is reapplied to the corrupted images. This corruption-based robustness measure follows the robustness evaluation framework of Hendrycks and Dietterich (2019), where common corruptions are used to assess robustness under unexpected image degradation. Moreover, robustness benchmarking for semantic segmentation tasks using corrupted data was previously demonstrated by Kamann and Rother (2020).

Relative Performance Drop (RPD). This metric calculates the percentage drop in mIoU, normalized by the original mIoU on the clean test set (Hendrycks & Dietterich, 2019; Kamann & Rother, 2020). Relative Performance Drop (RPD) is computed as $RPD = (mIoU_{clean} - mIoU_{corrupted}) / mIoU_{clean} \times 100$. RPD is self-normalized: each stage's corruption-induced drop is divided by that same stage's clean mIoU, so RPD measures retention relative to a stage's own clean performance. It is therefore distinct from the relative corruption degradation (rCD) of Kamann and Rother (2020), which normalizes corruption-induced degradation by a reference model's degradation; rCD is defined separately below. Lower RPD indicates better performance retention under the selected corruptions. RPD is given as a purely descriptive figure for each tested stage (E1 through E7) and for the SegNeXt-B teacher as a descriptive reference. RPD itself is not used as the target of an inferential test; the inferential robustness test uses per-image mIoU-C for E1 versus E6. It therefore is not presented as proof that agricultural field performance is also retained or that corruption robustness has a correlation with on-field behavior. The retention is demonstrated on the selected corruptions within the controlled testbed only.

$$RPD = \left(\frac{mIoU_{clean} - mIoU_{corrupted}}{mIoU_{clean}} \right) \times 100$$

Relative Corruption Degradation (rCD). Following Kamann and Rother (2020), rCD is reported as a descriptive metric that isolates corruption-induced degradation while controlling for clean-accuracy differences across stages. With $D = 1 - \text{mIoU}$ and using E1, the FP32 undistilled student, as the reference, rCD for a stage and corruption type is the sum over severity levels 1–3 of corrupted D minus clean D for that stage, divided by the corresponding sum for E1, reported per corruption and averaged across corruptions. By construction, rCD equals 1 for E1; a value below 1 indicates that a stage degrades less than the baseline student, while a value above 1 indicates greater fragility. Because the reference is internal rather than an external fixed architecture, rCD is a within-study relative measure and is not cross-study comparable. Per-corruption values are reported so that near-zero reference denominators on mild corruptions remain visible. rCD is descriptive only and is not a target of inferential testing; the SegNeXt-B teacher is excluded from rCD.

In addition to mIoU-C and RPD summaries given per stage, mIoU-C values are also broken down for each of the five corruption types (Gaussian noise, JPEG compression, motion blur, brightness, fog). For every corruption type, the mIoU-C score is averaged across the three severity levels (levels 1 to 3), reported for E1 through E7 and for the SegNeXt-B teacher where applicable. From these numbers, a descriptive table with five corruption rows and stage columns for E1 through E7 plus the teacher reference will be compiled, supporting qualitative robustness comparisons across corruption types and pipeline stages. The per-corruption analysis is descriptive only and will not include any form of statistical testing. Its goal is to point out which corruption types are most influential, highlight relative robustness patterns across stages, and form the structural basis for the robustness tables discussed later on.

4. Efficiency Metrics

Model size and parameter count. The model size describes the memory footprint of each model version. Parameter count is used to assess the architectural complexity of a model variant. Storing weights as 8-bit integers (INT8) instead of 32-bit floats saves roughly four times the storage per weight, because the 8-bit representation is four times smaller. INT8 QAT does not change the actual number of weights, as this is a feature of the model's architecture rather than the precision by which the weights are encoded. Thus, while file size between the FP32 configurations (E1, E2, E3) and INT8 configurations (E4, E5, E6, E7) may differ after successful conversion, the weight count remains identical for configurations using the same base student architecture. Model size is reported in two complementary forms. The primary, deterministic measure is the theoretical parameter-byte footprint, in which each weight in an INT8-quantized layer is counted as one byte, each weight retained in floating point as four bytes, and biases together with per-channel scales and zero-points as four bytes each; the FP32-to-INT8 compression ratio is computed from this footprint to avoid contamination by serialization-container overhead. As a practical supplement, the on-disk size of the serialized deployable artifact is reported and is measured on the same model instance used for CPU-proxy latency, so that all efficiency metrics describe a single artifact. Where backend operator support retains certain layers in floating point, the quantized and floating-point parameter partition is disclosed alongside the size figures, and any deviation from the nominal fourfold reduction is attributed to this mixed-precision composition.

Floating Point Operations (FLOPs). FLOPs measure the number of floating-point operations required for a single forward inference pass on one input image at 512×512 resolution. In this study, FLOPs are computed as MACs × 2 (the standard convention: one multiply-accumulate = one multiplication + one addition; Howard et al., 2019). FLOPs are profiled from the FP32 architecture using fvcare and reported descriptively.

$$1 \text{ MAC} \approx 2 \text{ FLOPs}$$

For a standard convolution, the approximate FLOPs may be expressed as:

$$\text{FLOPs} \approx 2 \times H \times W \times C_{in} \times C_{out} \times K^2$$

where H and W are the output feature map height and width, C_{in} and C_{out} are the input and output channels, and K is the kernel size. However, standard MAC computation is inapplicable to depthwise separable convolution, which MobileNetV3 employs and is a form of non-standard convolution layer type, also including inverted residual blocks and LR-ASPP/Lite-ASPP-style pooling branches. Thus, the study uses standard software profilers for models run with an input size of 512×512 using a framework profiler, `fvcore` as the sole FLOPs/MACs profiler, since they rely on layer-level calculation and account for the custom layer types more reliably (Howard et al., 2019). MACs are reported as the primary computation count, with FLOPs given using the convention $\text{FLOPs} = 2 \times \text{MACs}$. FLOPs are profiled from the FP32 model architecture and used as a descriptive efficiency indicator. All operation counts are obtained using `fvcore`. Values are reported primarily as multiply-accumulate operations (MACs), and when FLOPs are quoted, the convention $\text{FLOPs} = 2 \times \text{MACs}$ is used. The same tool and convention are applied to the SegNeXt teacher and the MobileNetV3 student so the teacher-student comparison uses one counting convention. Elementwise operations such as squeeze-and-excitation gating, Hardsigmoid/Sigmoid, residual adds, and bilinear upsampling are not counted under this convention and are reported as excluded from the theoretical MAC/FLOP total. INT8 QAT reduces model size and potentially inference time depending on backend and hardware support, but it does not change the theoretical FLOPs count for the given architecture.

Inference latency and peak memory usage. Both CPU proxy inference time and memory usage are measured on CPU devices (`torch.device('cpu')`) in PyTorch with a batch size of 1 using `torch.utils.benchmark`. The first 20 forward passes are treated as warm-up runs to allow the CPU to use caches normally. Subsequently, 100 measured forward passes are recorded to obtain the median latency and interquartile range. For the INT8 configurations, the quantized backend engine used for CPU-proxy latency is explicitly reported. If the RunPod CPU environment is x86, the PyTorch quantized backend is set to `torch.backends.quantized.engine = 'x86'` for latency measurement, because QNNPACK kernels are ARM-oriented and not representative on an x86 CPU proxy. For this x86 CPU-proxy measurement, a separate INT8 copy of each model is prepared with the `fbgemm/x86` quantization configuration (per-channel symmetric weights, `reduce_range = True`) and benchmarked on the x86 backend; the QNNPACK-configured model remains the artifact reported for accuracy and size, and any accuracy difference between the two quantization configurations is documented. On-device ARM/QNNPACK latency is not benchmarked. Median, interquartile range, and 95th-percentile latency are reported. Peak resident memory is measured on CPU via `resource.getrusage(ru_maxrss)` or `psutil` RSS and reported as a delta over the post-load baseline, defined as the resident set size recorded immediately after model construction and weight loading but before the first forward pass. CUDA-specific memory functions are not used, and `tracemalloc` is not used because it captures Python-level allocations but can undercount ATen tensor memory. Static model size is reported separately as the serialized deployable artifact size on disk. After each measurement, temporary tensors and model references are cleared where applicable, and the profiler state is reset if supported by the tool used. Each measurement for latency and memory usage is identified by the label CPU proxy. No tests are conducted on actual ARM devices or other embedded hardware. The

experiments, therefore, do not provide data on the potential inference speed for real, embedded edge devices.

5. Comparative statistical approach

All seven student stages (E1–E7) and the SegNeXt-B teacher reference are evaluated under identical experimental conditions; the resulting comparisons are tabulated in Table 3.6. To ensure a fair comparison and that any discrepancy observed in accuracy is attributable solely to changes in the models, distillation approach, or post-training quantization method rather than the dataset itself, the same dataset partitions, preprocessing methods, and evaluation metrics were kept unchanged across all tested models.

Table 3.6 Pairwise Ablation Comparisons and Statistical Treatment

Comparison	Investigative Purpose	Primary Variables Tested	Statistical Treatment	Hypothesis Tested
E1 vs. E2	Determine whether Logit Knowledge Distillation improves the FP32 student baseline	Per-image mIoU	Wilcoxon signed-rank, one-tailed (primary); paired t-test (sensitivity); Holm-Bonferroni correction across all 8 inferential tests; rank-biserial r_{rb} , Hodges-Lehmann shift, Cohen's d_z , mean $\Delta mIoU$, and BCa 95% CIs reported	H _{1a}
E1 vs. E3	Determine the total FP32 distillation gain from the undistilled	Per-image mIoU	Reported descriptively as the total FP32 distillation gain (E1 → E3): mean $\Delta mIoU$ and Hodges-Lehmann	Supporting comparison for

	baseline to the Logit KD + CWD student		shift with BCa 95% confidence intervals. Not included in the eight-test Holm-Bonferroni family, as this comparison is the composition of the E1 vs E2 and E2 vs E3 increments and corresponds to no independent hypothesis.	H _{1a} and H _{1b}
E2 vs. E3	Determine whether adding Channel-Wise Knowledge Distillation (CWD) improves the student beyond Logit KD alone	Per-image mIoU	Wilcoxon signed-rank, one-tailed (primary); paired t-test (sensitivity); Holm-Bonferroni correction across all 8 inferential tests; rank-biserial r_{rb} , Hodges-Lehmann shift, Cohen's d_z , mean $\Delta mIoU$, and BCa 95% CIs reported	H _{1b}
E4 vs. E5	Determine whether INT8 QAT preserves segmentation accuracy better than INT8 PTQ	Per-image mIoU	Wilcoxon signed-rank, one-tailed (primary); paired t-test (sensitivity); Holm-Bonferroni correction across all 8 inferential tests; rank-biserial r_{rb} , Hodges-Lehmann shift, Cohen's d_z , mean $\Delta mIoU$, and BCa 95% CIs reported	H _{1c}

E7 vs E6	Determine whether INT8 QAT preserves segmentation accuracy better than INT8 PTQ on the distilled student	Per-image mIoU	Wilcoxon signed-rank, one-tailed (primary); paired t-test (sensitivity); Holm-Bonferroni correction across all 8 inferential tests; rank-biserial r_{rb} , Hodges-Lehmann shift, Cohen's d_z , mean $\Delta mIoU$, and BCa 95% CIs reported	H_{1c} extended
E4 vs E7	Determine whether prior distillation (E3) improves the PTQ path compared with PTQ from the undistilled student (E1)	Per-image mIoU	Wilcoxon signed-rank, one-tailed (primary); paired t-test (sensitivity); Holm-Bonferroni correction across all 8 inferential tests; rank-biserial r_{rb} , Hodges-Lehmann shift, Cohen's d_z , mean $\Delta mIoU$, and BCa 95% CIs reported	Supporting comparison for H_{1d}
E5 vs E6	Determine whether prior distillation (E3) improves the QAT path compared with QAT from the undistilled student (E1)	Per-image mIoU	Wilcoxon signed-rank, one-tailed (primary); paired t-test (sensitivity); Holm-Bonferroni correction across all 8 inferential tests; rank-biserial r_{rb} , Hodges-Lehmann shift, Cohen's d_z , mean $\Delta mIoU$, and BCa 95% CIs reported	H_{1d}

E3 vs. E6	Determine how much segmentation accuracy is retained after applying INT8 QAT to the E3 distilled student (Logit KD + CWD)	Per-image mIoU (primary unit of paired testing); efficiency metrics (model size, FLOPs, latency, peak memory) and clean→corrupted comparison via RPD reported descriptively only	Wilcoxon signed-rank, one-tailed (primary); paired t-test (sensitivity); Holm-Bonferroni correction across all 8 inferential tests; rank-biserial r_{rb} , Hodges-Lehmann shift, Cohen's d_z , mean $\Delta mIoU$, and BCa 95% CIs reported; RPD, model size, FLOPs, latency, and memory are descriptive only	Accuracy retention check; supporting H _{1d}
E1 vs E6	Determine whether the full proposed pipeline (distillation then INT8 QAT) improves the original FP32 student baseline in overall accuracy–efficiency–robustness trade-off	Per-image mIoU (accuracy); per-image mIoU-C (robustness); efficiency metrics descriptive	Wilcoxon signed-rank, one-tailed on per-image mIoU (primary accuracy test); Wilcoxon signed-rank, one-tailed on per-image mIoU-C (the one inferential robustness test); paired t-test (sensitivity for both); Holm-Bonferroni correction across all 8 tests; rank-biserial r_{rb} , Hodges-Lehmann shift, Cohen's d_z , mean $\Delta mIoU$, and BCa 95% CIs; efficiency metrics descriptive only	H _{1d} (main pipeline evaluation; both accuracy on per-image mIoU and robustness on per-image mIoU-C are counted as two of the 8 inferential tests)

Teacher vs. E6	Evaluate the accuracy–efficiency–robustness trade-off of the final proposed pipeline against the high-capacity teacher reference	mIoU, Dice Coefficient, corrupted mIoU, RPD, model size, parameter count, FLOPs, inference latency, and peak memory usage	Descriptive comparison only. The teacher is not a deployment candidate and is not an inferential comparator. mIoU, mIoU-C, model size, FLOPs, latency, and memory are reported side-by-side as a descriptive upper-bound reference	Descriptive reference only
Published PlantSeg	Contextualize the final proposed model against	Published aggregate mIoU and mAcc values compared with	Descriptive external benchmark comparison only,	Descriptive context

benchmarks vs. E6	previously reported PlantSeg benchmark results where metric and protocol compatibility can be established	the proposed model's mIoU and related reported metrics	unless external baselines are reproduced under the same protocol	ual support only
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